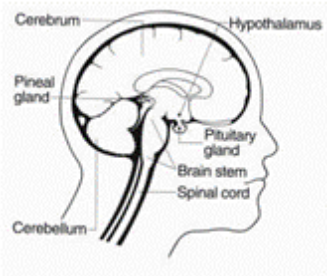


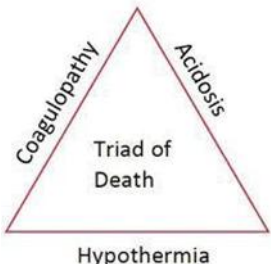
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ADULT – EMERGENCY SURGICAL CRICOTHYROTOMY**OVERVIEW:**

A cricothyrotomy is a surgical procedure to establish an emergency airway. It is an invasive procedure with multiple possible complications and should be performed only on patients that are at high risk of death if an immediate airway is not established. One must first consider/attempt all alternative airway measures (e.g. OPA/NPA, ET intubation (direct line of site or video laryngoscope), and Supraglottic device.

REQUIREMENTS:

- Be a licensed paramedic credentialed through the Deputy Chief of Training
- Be a paramedic with Sumner County EMS for at least 6 months.
- Must have completed bi-annual training sessions as required by Sumner County EMS

INDICATIONS:

- Unsuccessful attempts to visualize the airway by direct laryngoscopy or video laryngoscopy
- Inability to intubate and/or ventilate
- Severe facial trauma or total airway obstruction not relieved by other methods

COMPLICATIONS:

- Bleeding (can be severe)
- Misplacement (esophageal or soft-tissue placement)
- Damage to surrounding structures such as vasculature, vocal cords, esophagus or trachea
- Infection

CONTRAINDICATIONS:

- Given that you will only be performing this procedure on patients who have a very high probability of dying without it, most contraindications are therefore relative. The one absolute contraindication is pediatric patients ≤ 10 years old, as the available equipment is too large for this population. Pediatric patients may be treated under the pediatric emergency cricothyrotomy protocol. The following are examples of patients with a high risk of complications:
 - Age ≤ 10 years old (Refer to Pediatric Needle Cricothyrotomy Protocol)
 - Bleeding disorders
 - Massive neck swelling (blood/tumor) causing inability to identify landmarks

EQUIPMENT NEEDED	
Full BSI / PPE is recommended	Bougie stylet
Sharps container	6.0 ET Tube (other sizes may be used PRN)
Suction device	10 mL syringe
BVM with O2 supply	EtCO2 monitoring
Antiseptic / Betadine swabs	Securing device
#10 scalpel	Bandaging materials

CONTINUED ON NEXT PAGE

ADULT – EMERGENCY SURGICAL CRICOTHYROTOMY (Procedure)

EMR

- Obtain basic assessment
- Attempt to open airway manually with head-tilt chin-lift or jaw thrust
- If airway obstruction is visible and able to be reached, remove obstruction manually (No blind sweep)
- Attempt to ventilate with BVM

EMT

- Obtain vital signs including SpO₂, blood pressure, and ECG
- Attempt to establish a less invasive airway (Supraglottic device)

AEMT

- Prepare for waveform capnography monitoring and apply to ETT after placement
- Establish vascular access as indicated

PARAMEDIC

- Position patient supine and extend the neck as need to improve anatomic view
- Prep neck with antiseptic
- Using your non-dominant hand, stabilize the larynx and locate the following landmarks:
 - Thyroid cartilage (Adam's apple) and cricoid cartilage
 - The cricothyroid membrane lies between these cartilages
- Make an approximate 3 cm vertical incision 0.5 cm deep through the skin and fascia, over the cricothyroid membrane. Using a finger, dissect the tissue and locate the cricothyroid membrane
- Make an approximate 1.5 cm horizontal incision through the cricothyroid membrane
- With your finger, bluntly dilate the opening through the cricothyroid membrane
- Insert the bougie curved tip first through the incision angled toward the patient's feet
- Advance the bougie into the trachea feeling for "clicks" of tracheal rings and until resistance is met and it cannot be advanced any further
- Advance a 6.0 mm ET Tube (ensuring cuff is deflated) over the bougie and into the trachea
- Remove the bougie while stabilizing ET Tube, ensuring it does not become dislodged
- Inflate the cuff with 5-10 mL of air
- Confirm appropriate, tracheal placement with:
 - Symmetrical rise and fall of the chest wall
 - Auscultation of equal breath sounds and absent epigastric sounds while ventilating
 - Condensation in the ET Tube
 - Capnography
- Secure the ETT
- Bandage as necessary

ADULT – EMERGENCY SURGICAL CRICOTHYROTOMY (Procedure) (Continued)

- Frequently reassess placement of the ETT, especially after movement of patient
- Continuously monitor placement of ET Tube and ventilation status using waveform capnography

See “Post Intubation Management” section of “Facilitated Intubation and RSI” protocol to provide sedation and pain management.

FACILITATED INTUBATION AND RSI**OVERVIEW:**

Rapid sequence induction/intubation (RSI) is a series of maneuvers utilizing sedation and paralysis to establish an advanced airway in a critically ill patient. This is an advanced procedure with a potential for high-risk complications and should only be performed as an absolute life-saving procedure. It should only be performed after all other less invasive forms of airway control have been attempted or considered. At no time should a paramedic feel pressured to perform this procedure if he or she is not comfortable with its application on a given patient. ***All paramedics may initiate this procedure with the administration of choice sedatives (chemically facilitated intubation). However, you must meet the requirements below in order to perform full RSI, to include administration of paralytics.***

REQUIREMENTS TO PERFORM RSI (Use of paralytics):

- Be a licensed paramedic and employee of Sumner County EMS for at least 6 months, or experienced personnel *may* be approved by the Medical Director
- Be in good standing with the service regarding clinical issues
- Complete initial AND bi-annual airway, RSI and cricothyrotomy training courses

INDICATIONS:

To establish an airway in a patient who is at risk of death secondary to loss of airway or inability to ventilate, and the airway cannot be controlled by conventional means. Examples of patients in which pre-hospital RSI might be indicated include, but are not limited to the following:

- Facial or head trauma patients with loss of airway control
- Severe respiratory distress with hypoxia and/or respiratory exhaustion
- Burn patients with airway involvement and respiratory distress
- Overdose with loss of airway protection and hypoxia

CONTRAINDICATIONS:

- Allergy to any one of the agents

CONTRAINDICATIONS TO SUCCINYLCHOLINE:

- History of malignant hyperthermia
- Renal failure
- Spinal cord injury greater than 24 hours old or neuromuscular disease
- Severe burns greater than 8 hours old
- Massive crush injuries
- Pesticide poisoning
- Penetrating eye injuries

CONTINUED ON NEXT PAGE

FACILITATED INTUBATION AND RSI (Procedure)

EMR

- Position patient and open airway manually with head-tilt chin-lift or jaw thrust maneuver
- Insert OPA as indicated
- Preoxygenate by administering 100% oxygen and ventilate with BVM
- Obtain basic assessment including manual blood pressure

EMT

- Obtain vital signs including SpO2
- Place electrodes on patient to obtain ECG
- Attempt less invasive airway device (NPA, OPA, or Supraglottic Device)
- Place ETCO2 NC on patient at 15 LPM to begin hyperoxygenation

AEMT

- Begin capnography monitoring using ETCO2 NC
 - Note that ETCO2 measurement may be inaccurate while using high-flow oxygen >5 LPM
- Obtain vascular access
- If hypotensive, begin fluid resuscitation as indicated

PARAMEDIC

- Give **SEDATIVE** (Induction) – Use appropriate, available induction agent:
 - Ketamine 1-2 mg/kg IV/IO
 - First choice induction agent (sedative)
 - Avoid if patient is hypertensive, or in presence of tachyarrhythmias or acute MI
 - Etomidate 20 mg IV/IO, when Ketamine is contraindicated (**first alternative to Ketamine**)
 - Versed 5 mg IV/IO (**only used as an alternative to Etomidate**)

*** CONSIDER ATTEMPTING FACILITATED INTUBATION AFTER AVAILABLE INDUCTION AGENT IS GIVEN ***

**** REFER TO CONTRAINDICATIONS FOR SUCCINYLCHOLINE ON PREVIOUS PAGE PRIOR TO GIVING! ****

- Give PARALYTIC (short-acting)
 - Administer Succinylcholine 1 mg/kg IV/IO (**Standard Adult dose: 100 mg**) or Rocuronium 1 mg/kg IV/IO (**Standard Adult dose: 100 mg**)

CONTINUED ON NEXT PAGE

FACILITATED INTUBATION AND RSI (Continued) – POST-INTUBATION MANAGEMENT

PARAMEDIC

- Placement and Proof
 - Intubate when patient becomes flaccid, often after fasciculations. If the patient cannot be intubated after 2 attempts, then use an alternative airway such as the iGel or basic airway adjuncts and continue to ventilate patient until the Succinylcholine wears off
 - Confirm placement with capnography / capnometry, auscultation, etc.
 - **(Minimum of 5 verification techniques)**

For any patient who is already intubated (such as interfacility transfers), ALL PARAMEDICS can begin with this section to maintain sedation. Go to long-acting paralytics as needed, seen below.

- Post-Intubation Management
 - Administer 100% Oxygen, and titrate to > 94% when possible
 - Maintain ETCO₂ of 35-45 mmHg when possible
 - Secure ET Tube
 - Avoid delivering excessive volumes / pressures with the BVM to prevent risk of barotrauma
 - Use a PEEP valve (5-10 cmH₂O) in patients with a stable blood pressure
- Give maintenance **SEDATION** and **PAIN CONTROL**. This **MUST BE DONE!**
 - Ketamine 0.5-1 mg/kg IV/IO, one time dose, if Ketamine was used for induction prior to RSI
 - Versed 2-5 mg IV/IO
 - Repeat 2-5 mg as necessary and titrate to desired effect
 - Fentanyl 100 mcg IV/IO may also be given
 - Repeat as necessary and titrate to desired effect
- Administer long-acting paralytic after correct placement is assured
 - Rocuronium (Zemuron) 1 mg/kg (**Standard Adult dose: 100 mg**)
 - Actual dose range is 0.6-1 mg/kg. May give 1 mg/kg for ease of administration.
 - Norcuron (Vecuronium) 0.1 mg/kg IV/IO, (**Standard Adult dose: 10 mg**), if Rocuronium is unavailable.
- Transport without delay

PUSH-DOSE PRESSORS**INDICATIONS:**

- Post-ROSC Hypotension
- RSI / Management of peri-intubation hypotension
- Bridge to continuous Epinephrine infusion
- Management of specific shock states including:
 - Anaphylactic Shock
 - Septic Shock (After 30 mL/Kg fluid bolus with ongoing signs and symptoms of shock)
 - Neurogenic Shock
 - Cardiogenic Shock
 - Hypovolemic Shock (Refractory to 20 mL/Kg of fluid resuscitation)

CONTRAINDICATIONS:

- Tachydysrhythmias
- Hypotension responsive to fluid resuscitation

PRECAUTIONS:

- May cause or worsen tachycardia / tachydysrhythmias
- May cause acute hypertension, and thus possibly reflex bradycardia

MIXING INSTRUCTIONS:

- Take a 10 mL syringe filled with 9 mL of normal saline. If premade saline syringes are available, eject 1 mL of saline from the 10 mL saline syringe, leaving 9 mL remaining
- Into this syringe, draw up 1 mL of Epinephrine from the cardiac Epinephrine ampule
 - Cardiac amp contains 10 mL of Epinephrine with a concentration 100 mcg/mL or 1:10,000)
- Shake well. You now have 10 mL of Epinephrine 10 mcg/mL (1:100,000)
- Confirm medication and concentration with partner
- **Label the syringe**

EMR

- Obtain basic assessment including manual blood pressure

EMT

- Obtain vital signs including SpO₂, blood pressure, and ECG

AEMT

- Establish vascular access and initiate fluid resuscitation as indicated
- Initiate waveform capnography monitoring, if indicated

CONTINUED ON NEXT PAGE

PUSH-DOSE PRESSORS (Continued)

PARAMEDIC

- Administer 5-10 mcg (0.5 mL – 1 mL) doses via rapid IV push every 2-5 minutes as needed to achieve hemodynamic goal (Systolic BP >90 mmHg and/or MAP $\geq$ 65 mmHg)

OROGASTRIC (OG) TUBE INSERTIONINDICATIONS:

- To facilitate gastric decompression of the intubated patient after endotracheal intubation
- For use with iGel Supraglottic Airway
- Submersion injuries (near-drowning)
- Intubated Pediatric Patients
- Any patient who has been aggressively artificially ventilated

CONTRAINDICATIONS:

- Known esophageal disease (Such as esophageal varices)
- Previous gastric bypass surgery
- Penetrating neck trauma

EQUIPMENT:

1. 18 French Sump Catheter
2. Lubricant (water-soluble)
3. 60 mL Catheter-tip Syringe
4. Tape

Special Note:

- If you cannot place the OG tube quickly (no more than 2 attempts), forego the procedure – DO NOT delay transport.
- The blue air vent must remain patent to ensure proper sump function and the prevent damage to the gastric lining during suction.

CONTINUED ON NEXT PAGE

OROGASTRIC (OG) TUBE INSERTION (Procedure)-(Continued)

PARAMEDIC

PROCEDURE:

1. Estimate the length of the tube needed to reach the stomach by measuring the tube from the corner of the mouth to the earlobe and down to the xiphoid process. Mark the length with tape.
2. Lubricate the Salem Sump tube with a water-soluble lubricant (e.g. KY Jelly).
3. Insert the tube through the oropharynx or through the gastric access lumen on the iGel Airway until the marked depth is reached.
4. If the tube coils in the posterior pharynx, direct laryngoscopy can be utilized to place the tube in the esophagus. If the patient begins to cough, you may be advancing into the trachea. Reattempt insertion.
5. Verify placement (see OG Placement Verification).

PLACEMENT VERIFICATION:

1. Using a 60 mL catheter tip syringe, instill 30 mL of air into tube and auscultate over epigastrium for air sounds.
2. Aspirate for gastric contents and assess for cloudy, green, tan, brown, bloody or off-white color contents consistent with gastric contents.
3. Secure tube with tape.

GASTRIC DECOMPRESSION:

1. Attach the tube to suction (and suction intermittently as needed). You can also provide continual suction by setting the suction device at 20-60 mmHg negative pressure (TORR).
2. If suction is not readily available, connect the 60 mL syringe to the tube, keeping the (blue) air vent patent. This will allow the sump function of the tube to work until suction can be applied and will prevent gastric contents from leaking from the tube.

PAIN MANAGEMENT (ACUTE TRAUMATIC INJURIES)

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of patients with a chief complaint / signs and symptoms of an acute traumatic injury.

EMR

- Assure ABC's are intact.
- Administer Oxygen as needed.
- Obtain manual blood pressure.

EMT

- Initiate pulse oximetry and waveform capnography monitoring.
- Initiate cardiac monitoring at the discretion of the Paramedic.

AEMT

- Establish IV access or IO access if needed for a critical patient.
- If there is an obvious fracture, refer to the "Fractures - General Care" protocol.

PARAMEDIC

- Consider Toradol (Ketorolac) 15 mg IV or 30 mg IM (DO NOT Give in patients > 65 years old) for the following:
 - Isolated Musculoskeletal injuries.
 - Kidney Stones.
 - Back pain.
- No standing orders for repeat doses of Toradol
- For acute traumatic injuries in the absence of hypotension and altered mental status (AMS) due to head injury, administer: Fentanyl 1.0 mcg/kg slow IV/IO
 - Max initial dose of 50 mcg in geriatric patients.
 - In the presence of hypotension or AMS due to head injury, consult with on-line medical control prior to giving any analgesics.
 - If patient is allergic to Fentanyl, or if Fentanyl is unavailable, consider:
 - Morphine 0.05 – 0.1 mg/kg IV/IO/IM, max initial dose of 5 mg in geriatrics
- If unable to obtain vascular access, give either Fentanyl or Morphine IM once.

Contraindications for Toradol

- Anticoagulants / blood thinners of any type
- < 16 years of age or > 65 years of age
- Major trauma or head injuries
- Chronic kidney disease
- Allergies to NSAIDS

CONTINUED ON NEXT PAGE

PAIN MANAGEMENT (ACUTE TRAUMATIC INJURIES) - (Continued)

- If pain is not relieved with Morphine or Fentanyl alone, then consider giving Ketamine 20 mg IV/IO.
 - The typical supply of Ketamine comes 500 mg/10 mL, so 20 mg would be drawn up at 0.4 mL.
 - To be given after initial Fentanyl or Morphine dose (not given by itself for pain management).
 - Standing orders for Ketamine are for adults only.
- Fentanyl may be repeated at 0.5 mcg/kg increments, titrated to effect, given at least 5 minutes apart.
 - Max 2nd dose of 25 mcg in patients > 65 years' old.
- Morphine may be repeated one time if needed, 5 minutes after the initial dose.
 - Max repeat dose of 2 mg in geriatric patients.
- Use extreme caution when administering narcotics to geriatric population (≥65 years of age).
- Transport as indicated – MUST be transported ALS with Paramedic providing care.
- THERE ARE NO STANDING ORDERS FOR ANALGESICS IN ABDOMINAL PAIN, OR OTHER MEDICAL COMPLAINTS. CONSULT ON-LINE MEDICAL CONTROL AS NEEDED.

INTRAOSSIOUS (IO) INFUSION (Procedure)

AEMT

PARAMEDIC

Approved Sites for AEMTs or Paramedics, **only use sites you are trained in identifying:**

Adults – Proximal tibia or humeral head. Humeral head may deliver a faster rate of infusion in critical shock or trauma patients who are in need of emergent fluid administration.

Pediatrics – Proximal tibia will be the preferred site, and humeral head access may be utilized for patients ≥ 8 years of age.

Indications:

- Any patients who are in need of emergency vascular access where peripheral IV access is difficult
- IO may be the initial vascular access attempted in critical (non-responsive to pain) pediatric patients
- IO may be the initial vascular access attempted in adult cardiac arrest patients. However, peripheral IV access still remains an industry standard initial vascular access (EJ access is considered peripheral)

Contraindications:

- Fracture in target bone
- Previous, significant orthopedic procedures at insertion site, prosthetic limb or joint
- IO access (or attempted IO access) in the targeted bone within the past 48 hours
- Infection at the area of insertion
- Excessive tissue (severe obesity) and/or absence of adequate anatomical landmarks

Procedure:

1. Gain consent, inform and explain the need for IO to the conscious patient who requires the IO access
2. Prepare the appropriate site with the proper placement of the extremity, select the proper needle size
3. Apply and maintain aseptic technique throughout the procedure
4. Properly insert the IO in the target bone/site. Note that the inability to aspirate (withdraw) bone marrow does not necessarily mean insertion was unsuccessful. Verify placement with the aspiration of marrow if possible.

---(Go to step 6 if unresponsive) ---

5. **For patients responsive to pain, administer Lidocaine 2% (anesthetic) ASAP following IO insertion**
 - ADULT – Administer 40 mg slow push, over 2 minutes
 - PEDIATRICS – Administer 0.5 mg/kg (up to a max of 40 mg) slow push, over 2 minutes
 - Allow initial doses of Lidocaine to dwell in the site for 1 minute before flushing
 - Provide a rapid (steady) flush with NS, 10 ml flush in adults, 5 ml flush in pediatrics
 - After the initial flush, **repeat** Lidocaine 2% at HALF the initial dose and allow to dwell again for 1 minute
 - (Repeat doses of Lidocaine 2% would be 20 mg in adults or 0.25 mg/kg in peds)
 - Proceed with fluids or medications administration

CONTINUED ON NEXT PAGE

INTRAOSSEOUS (IO) INFUSION (Procedure) – (Continued)

6. If the IO will not flush or flow, it may be against the inner wall inside the medullary space. Consider retracting the needle very slightly to clear the distal end from against the inside of the bone and attempt to flush again
7. **WARNING** - When using a pressure bag or the other pressure infusion device, be careful not to allow air from the IV bag to enter the drip chamber of the tubing and inadvertently be infused into the patient
8. Secure the IO needle at the site using tape, specifically designed devices, or other bulky dressings
9. Observe for any signs of swelling with infusion or other indication of non-patency
10. If required, IO needles can be removed by attaching a 10 mL syringe to the hub of the IO needle and pulling out with a gentle twisting motion. Cover the site with a dry sterile dressing and secure as indicated

Note that any medication that is given via IV may also be given via the IO route as well.

VENTILATOR PROTOCOL (ADULTS ONLY)

PARAMEDIC

INDICATIONS:

1. Patient must be intubated
2. Transfers
3. Prolonged transports
4. CPAP (See "CPAP" protocol)
5. BiPAP (See "BiPAP" protocol)

CONTRAINDICATIONS:

1. Endotracheal tube not in position and confirmed
2. Pulseless patient/code situation
3. Brief transport/delay in transport

QUALIFICATIONS/TRAINING REQUIREMENTS:

1. Paramedics with at least 2 years of experience that have completed the state mandated Ventilator Class
2. Renewal requires a 2-hour ventilator update class annually

INITIAL SETTINGS:

IF PATIENT IS TRANSFERRED FROM A HOSPITAL ON A VENT, USE THE SAME SETTINGS THE HOSPITAL IS USING.

1. Assist Control
2. Rate – 12
3. Tidal Volume – 400 mL or 5-8 mL/kg of lean body mass
4. FiO₂ – 100%
5. PEEP – 5

REQUIRED MONITORING:

1. Cardiac Monitoring
2. SpO₂ Monitoring
3. End-Tidal CO₂ Monitoring
4. Blood Pressure every 5 minutes
 - *On prolonged transfers of hemodynamically stable patients, blood pressure can be done as indicated*

VENTILATOR ADJUSTMENTS:

1. PEEP
 - If the patient is hypotensive, then PEEP = 0
 - If O₂ Sat < 90% and systolic blood pressure is > 100, may increase PEEP by 5
2. FiO₂ – May wean to keep O₂ Sats > 90%, but less than 100%
3. I:E ratio – Change to 1:4 for COPD or Asthma patients

DISCONTINUATION OF VENTILATOR:

IF AT ANY TIME THE PATIENT DEVELOPS PULMONARY OR HEMODYNAMICALLY COMPROMISE, TAKE THE PATIENT OFF THE VENTILATOR AND VENTILATE WITH A BVM AND 100% OXYGEN!

If current sedation is ineffective, and there is risk of failure to maintain endotracheal tube placement in the patient, **consider physician consult to seek orders** to discontinue current sedation, and default to our standing orders on page 7 of this section. Continue from that point, to possibly include long-acting paralytics. Remember, you MUST also be sure to give sedation as well (Versed, Fentanyl).

BILEVEL POSITIVE AIRWAY PRESSURE (BiPAP)

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of patients in respiratory distress to prevent respiratory failure.

INDICATIONS:

- Any patient who is in respiratory distress for reasons other than trauma or pneumothorax.
- Awake and able to follow commands.
- Over 12 years of age and able to fit the BiPAP mask.
- Have the ability to maintain an open airway.
- Systolic greater than 90 mmHg.
- Use of accessory muscles during respirations.
- Signs and symptoms consist with asthma, COPD, pulmonary edema, CHF or pneumonia.
- Pulse oximetry less than 94% with the listed indications above.

CONTRAINDICATIONS:

- Respiratory arrest or Apneic.
- If the patient has a tracheostomy in place.
- Active vomiting or has upper GI bleeding.
- If the patient has decreased cardiac output

PROCEDURE:

EMR

- Assess / Stabilize ABC's as needed
- Attached pulse oximetry
- Administer O2 as needed

EMT

AEMT

PARAMEDIC

1. EMT's, AEMT's and Paramedics can apply BiPAP, but only Paramedics can monitor the patient after application.
2. Gather the appropriate equipment.
3. Assess vital signs and attach pulse oximeter.
4. Make sure the patient does not have signs/symptoms of a pneumothorax.
5. EXPLAIN THE PROCEDURE.
6. Connect the BiPAP to the oxygen outlet and place an end-tidal CO2 nasal cannula or place the ET tube Capnography between mask and device.
7. Ensure adequate oxygen supply to ventilate device.

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BILEVEL POSITIVE AIRWAY PRESSURE (BiPAP) - (Continued)

8. Select the sealing face mask and ensure the mask fits comfortably, seals the bridge of the nose, and fully covers the nose and mouth.
9. Secure the mask to the patient with provided straps or other provided devices.
10. Prior to setting the pressure always observe that the airway pressure gauge needle indicator is at the zero position.
11. If the patient is in moderate or severe distress adjust BiPAP to 10/6 cmH₂O and titrate to the best effect with a maximum of 17/6 cmH₂O. If more pressure is needed then contact medical control.
12. Check for air leaks by watching movement of the needle on the manometer gauge.
13. Monitor and document the patient's respiratory response to the treatment.
14. Continue coaching the patient and readjust the mask as needed.
15. Evaluate vital signs every 5-10 minutes.
16. If the patient deteriorates, remove the device and consider positive pressure ventilation. Intubate if indicated.

SPECIAL NOTES:

- If a nebulizer treatment needs to be given, the device must be changed over to CPAP, and the nebulizer "T" piece can be placed between the mask and the device.
- Documentation must be noted.
- For hospital transfers, if the patient is currently on a BiPAP transport ventilator, and the paramedic is ventilator certified, then the paramedic can choose to maintain the patient on a transport ventilator. If the paramedic is NOT ventilator certified, and the settings do not exceed the maximum settings for the disposable BiPAP device, then transfer the patient over to the disposable BiPAP device, make sure the patient tolerates it, and then transport.

VENTILATOR BIPAP (ADULT ONLY):

- Indications
 - In-hospital patient receiving Bi-PAP
 - Acute Respiratory Failure
 - Non-intubated patient requiring pressure support
 - Respiratory muscle exhaustion
- Contraindications
 - Intubated Patient or unable to maintain airway
 - Hypotension or hemodynamically compromised
 - Altered Mental Status or decreased LOC
 - Inability to maintain adequate mask seal
 - Facial trauma or abnormalities / deformities
 - High-risk of aspiration or excessive secretions

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BILEVEL POSITIVE AIRWAY PRESSURE (BiPAP) - (Continued)

- Quals / Training Requirements
 - Paramedics with at least 2 years of experience and have completed the state mandated Ventilator class
 - Renewal requires a 2-hour ventilator update class annually.

PARAMEDIC

- Discontinuation of Bi-PAP for pulmonary or hemodynamically compromised patients:
 - Take patient off of the ventilator and ventilate using a BVM and 100% O₂
- Initial Settings:
 - Use hospital settings is available
 - If not, revert to the normal initial settings:
 - Select Non-Invasive CPAP mode on ventilator
 - Set Inspiratory PAP (IPAP) to 10 cmH₂O
 - Set Expiratory PAP (EPAP) to 5 cmH₂O
 - FiO₂ – 100% then titrate down to minimum required to maintain >92% SpO₂
- Required monitoring:
 - Cardiac Monitoring
 - SpO₂ Monitoring
 - EtCO₂ Monitoring
 - Mental Status Monitoring
 - Blood pressure checked every 5 minutes
 - On prolonged transfers of hemodynamically stable patients the blood pressure may be assessed at 5-15min intervals as indicated.
- Ventilator adjustments:
 - IPAP – increase in 2 cmH₂O increments, Max of 20 cmH₂O
 - EPAP – Try to maintain 5 cmH₂O unless persistent hypoxia is observed, then increase in 2 cmH₂O increments, Max of 10 cmH₂O.
 - Ensure that the level of pressure support (IPAP-EPAP=PS) remains constant, and an adequate tidal volume.
- If patient is unable to tolerate BiPAP after being changed over to the transport ventilation but is maintaining equal and adequate ventilation and oxygenation, and patient comfort cannot be established by adjusting settings, consider consulting with the sending physician for on scene administration of an anxiolytic or mild sedative prior to departure.

CHEST FINGER THORACOSTOMY**OVERVIEW:**

- Chest Finger Thoracostomy is an action that is taken to make surgical incisions into a patient's chest to access the thoracic cavity in the event of traumatic death that may have resulted from a life-threatening tension pneumothorax. Then inserting the finger of the paramedic into the patient's chest cavity area to relieve pressure from air or fluid (blood) that has accumulated in the thoracic cavity.

REQUIREMENTS TO PERFORM PROCEDURE:

- Must be a paramedic
- Must be checked off by your Medical Director or Training Officer, and have paperwork on file to confirm competency.

INDICATIONS:

- Traumatic cardiac arrest from blunt force trauma or penetrating injuries to the thoracic cavity.

CONTRAINDICATIONS:

- Not checked off on the procedure
- If patient is conscious and not in traumatic arrest.

PROCEDURE:**PARAMEDIC****Paramedic – Standing Order with Special Credentialing**

1. Position the arm in an abducted manner to easily locate the fifth rib along the midaxillary line, falling within the "triangle of safety." Mark this specific area for incision.
2. Sterilize the designated incision area using an antiseptic swab stick to minimize the risk of infection.
3. Carefully execute a transverse incision, approximately 1-2 inches (3-5 cm) in length, over the marked location on the fifth rib. The incision should be just anterior to the midaxillary line.
4. Utilize large forceps to quickly dissect the tissue above the rib, extending through the intercostal muscles.
5. Penetrate the pleura using the forceps. Once in position, expand the forceps to create a sufficient opening
6. Retract the expanded forceps from the chest, ensuring the created tract remains open.
7. Insert a finger along the dissected tract to reach the pleural cavity. Perform a sweeping motion to widen the opening.
8. Evaluate for any release of air or blood from the newly created opening, as this will guide further medical intervention.
9. For documentation and potential future investigations, encircle each incision with a surgical marker. Label these incisions as either "EMS-R" for the right side or "EMS-L" for the left side to identify procedures performed by EMS.

LATERAL PATELLAR REDUCTION

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of patients with a chief complaint of a patellar dislocation.

Assessment/Indications

- Severe Knee Pain
- Lateral displacement of the patella
- Inability to straighten extremity
- Must be ≥ 10 years of age

EMR

EMT

- Assess ABCs and Stabilize as needed
- Oxygen and airway maintenance appropriate for the patient's condition
- Obtain vital signs
- Pulse oximetry
- If BLS ambulance, consider ALS Intercept

AEMT

- Establish IV / Vascular access
- Obtain EtCO₂ reading

PARAMEDIC

- Cardiac monitor
- Fentanyl 1 mcg/kg IV to a maximum of 100 mcg.
- Explain the procedure to the patient.
- Refer to next page for the procedure to reduce the patella.

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LATERAL PATELLAR REDUCTION – PROCEDURE (CONTINUED)

PARAMEDIC

1. Stand on the lateral side of the leg on which the patellar reduction is to be completed.
2. Slightly flex the injured leg at the hip to decrease tension on the quadricep muscles.
3. Extend the knee while applying gentle anteromedially directed force on the lateral patellar edge to lift the patella over the femoral condyle.
4. When the reduction is completed, reassess pulse, motor and sensation distally, then immobilize the knee so that the knee is in full extension.
5. Transport to the ED.

ADULT (ACLS) CARDIAC GUIDELINES

Any state licensed EMR, EMT, AEMT, Paramedic, may perform skills and assessment outlined in this protocol at or below their appropriate level.

AEMT level may give Epinephrine 1:10,000 1 mg IV or IO in cardiac arrest situations under the direction of a paramedic.

- In order to be able to administer this medication, the AEMT must be checked off by their training officer and paperwork, including the TN State EMS form, must be on file if the AEMT received their AEMT licensure prior to January 1, 2023.

EMR

- Always treat the patient, not the monitor.
- Cardiac arrest caused by trauma is treated by correcting the underlying problem.
- Protocols for cardiac arrest situations, presumes that the condition under discussion continually persists, the patient on a cardiac monitor remains in cardiac arrest, and CPR is being performed.
- Chest compressions and defibrillation are more important than the administration of medications or establishment of an advanced airway.
- All attempts should be made to minimize interruptions in compressions which include allowing no more than 10 seconds for pulse check or no more than 5 seconds to deliver breaths. If an advanced airway is in place, CPR is not stopped to deliver breaths.
- Attempts should be made to use the mechanical CPR device.
- The cardiac monitor and other necessary equipment shall be taken to the immediate side of any unconscious, known cardiac arrest, or possible cardiac arrest patients. Perform a "quick-look" by applying pads and assessing the underlying rhythm.
- Patients presenting in cardiac arrest should receive ACLS care with an emphasis in high quality CPR and early defibrillation prior to being moved to an ambulance. Any scene safety concerns are justification to modify this to keep EMS personnel as safe as possible.

EMT

- Pulse oximetry
- Glucose check
- If BLS ambulance, consider ALS Intercept

AEMT

- Establish IV / Vascular access
- Obtain EtCO₂ reading
- IV or IO access is the preferred method of delivery of drugs. If an IV cannot be started in an arrest patient IMMEDIATELY, start an IO line.
- **IO may be attempted initially in cardiac arrest patients. *Peripheral access should still be considered.*** Paramedics may perform EJ access, but no IV catheter larger than an 18 gauge may be used.
- AEMT's can administer 1 mg of Epinephrine 1:10,000 IV or IO under the direction of a paramedic.

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ADULT (ACLS) CARDIAC GUIDELINES (Continued)

PARAMEDIC

- After each IV medication, give a 20 to 30 mL bolus of IV fluid and elevate the extremity.
- The fluid of choice for administering medications to the patient in cardiac arrest is Normal Saline.
- Induced Hypothermia is no longer a protocol.
- Routine use of Sodium Bicarb in cardiac arrest may not be warranted until ROSC is obtained.
- Consider reversible causes for PEA / Asystole:
 - Hypoxia
 - Toxins
 - Hypovolemia
 - Tamponade (cardiac)
 - Hydrogen ion (acidosis)
 - Tension Pneumothorax
 - Hypo- / Hyperkalemia
 - Thrombosis (coronary, pulmonary)
 - Hypothermia
 - Trauma
- For example:
- If possible pre-existing acidosis, consider giving Sodium Bicarbonate.
- If renal failure w/ possible increased potassium or Calcium Channel Blocker OD, consider Calcium Chloride.
- If tension pneumothorax, needle decompression.
- If hypovolemia, Lactated Ringers bolus.
- If tricyclic overdose, give Sodium Bicarbonate. If narcotic overdose, give Narcan.
- 12-lead ECGs should be obtained on the scene for the following cardiac problems and then every 10 minutes until arrival at receiving facility:
 - Non-traumatic chest pain greater than 25 years of age.
 - Symptomatic tachyarrhythmia's such as wide-complex tachycardia's, atrial fibrillations, SVT or frequent ectopy.
 - Symptomatic bradycardias or heart blocks (Type I and II and AV dissociation).
 - Congestive Heart Failure / Pulmonary edema.

12-LEAD PLACEMENT GUIDELINES

*****Transmit all 12-Lead ECG'S that are concerning for possible MI*****

*****Upload ALL 12-Lead ECGs to PCR*****

Out of hospital 12-Lead ECG's and advance notification to the receiving facility speeds the diagnosis, shortens the time to fibrinolysis or catheterization, and may be associated with decreased mortality rates. Providers shall complete training for 12-Lead ECG's acquisition prior to utilizing protocol and ECG machines.

INDICATIONS:

- 12-Lead ECGs should be done on the scene for the following and then repeat every 10 minutes until arrival to the receiving facility:
 - Non-traumatic chest pain greater than 25 years of age.
 - Symptomatic tachyarrhythmia's such as wide-complex tachycardia's, atrial fibrillations, SVT or frequent ectopy.
 - Symptomatic bradycardias or heart blocks (Type I, II, and AV dissociation).
 - Congestive Heart Failure / Pulmonary edema.
 - CVA / Stroke.
 - Dyspnea, to rule out cardiac causes of breathing problems in adults.
 - Syncope (fainting) or near syncope – maintain high index of suspicion for underlying ACS/MI.
 - Patients who are likely to have atypical (silent) MI, females >45 yrs of age, diabetics, vague complaints of nausea, exertional dyspnea or other signs of underlying possible ACS.

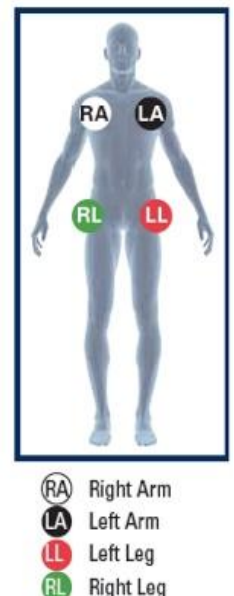
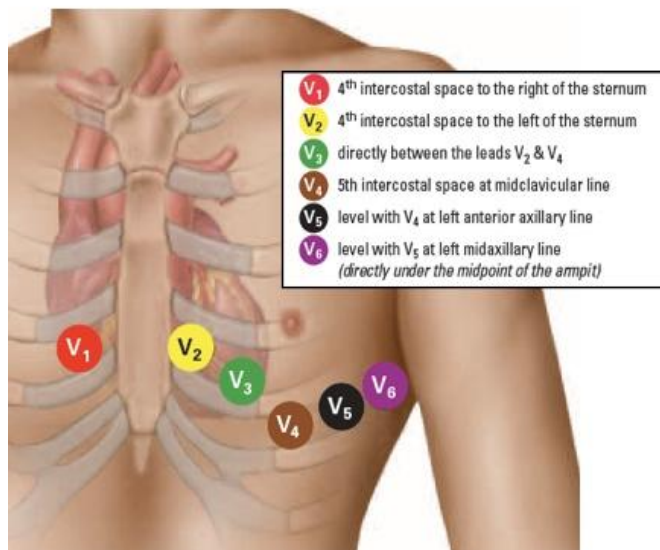
PRECAUTIONS:

- Ideally, 12-Lead ECG acquisition and treatment should occur concurrently but ultimately should not delay treatment of any life-threatening conditions:
 - Lethal dysrhythmias.
 - Respiratory emergencies.
 - Treatments such as O2, Aspirin, and NTG.
 - Request for advanced life support.
- Scene time should not be prolonged by acquisition of ECG.
- Factors that can reduce quality of tracing include dirt, oil, sweat, and other material on the skin and patient/vehicle movement.

"ACS" – Acute Coronary Syndrome – includes MI, STEMI, or PE cases.

EMTs and AEMTs may acquire and transmit 12 lead ECGs. However, only paramedics may interpret ECGs and provide care specific to these findings.

- TN EMS State Rules and Regulations approved.



12-LEAD ECG PROCEDURE (Continued):

1. Prepare all the equipment and ensure the cable is in good repair. Check to make sure there are adequate leads and materials for prepping the skin.
2. Prep the skin by first drying sweat or water. Lightly buff the electrode placement areas with an alcohol prep if the skin is dirty.
3. Place the four limb leads in accordance with the manufacturer's recommendations. Limb Lead electrodes are typically placed on the deltoid area and the lower leg or thigh. Avoid placing limb lead over body prominences.
4. Place the precordial leads (chest or V leads) in accordance with the manufacturer's recommendations. Proper placement is important for accurate diagnosis. Lead locations are identified as V1 through V6.
5. Locating the V1 position is critically important because it is the reference point for locating the placement of the remaining V leads. To locate the V1 position:
 - Place your finger at the notch in the top of the sternum.
 - Move your fingers slowly downward about 1.5 inches until you feel a slight horizontal ridge of elevation. This is the Angle of Louis where the manubrium joins the body of the sternum.
 - Locate the second intercostal space on the patient's right side, lateral to and just below the Angle of Louis.
 - Move your finger down two more intercostal spaces to the fourth intercostal space which is the V1 position.
6. Place V2 by attaching the positive electrode to the left of the sternum at the further intercostal space.
7. Place V4 by attaching the positive electrode at the midclavicular line at the fifth intercostal space. Note: V4 must be placed prior to V3.
8. Place V3 by attaching the positive electrode in the line midway between lead V2 and V4.
9. Place V5 by attaching the positive electrode at the anterior axillary line as the same level as V4.
10. Place V6 by attaching the positive electrode to the midaxillary line at the same level as V4.

CAUTION!

- NEVER USE THE NIPPLES AS REFERENCE POINTS FOR LOCATING THE ELECTRODES FOR MALE OR FEMALE PATIENTS BECAUSE NIPPLE LOCATION VARY WIDELY.
- WHEN PLACING ELECTRODES ON FEMALE PATIENTS, ALWAYS PLACE LEADS V3 THROUGH V6 UNDER THE BREAST RATHER THAN ON THE BREAST.

11. Ensure that all leads are attached.
12. Turn on the monitor.
13. Record the tracing by following the machine's specific acquisition procedure and function.
14. Document on the tracing, the patient's name, date and time the tracing was obtained.
15. Refer to the ST-ELEVATION MYOCARDIAL INFARCTION (STEMI) TRIAGE as indicated.

12-LEAD ECG PROCEDURE (Continued):**CONSIDERATIONS:**

- Acquire an additional 12-Lead ECG every 10 minutes or if the patient's clinical condition changes.

Personnel who are staffed as AEMTs on a BLS unit are encouraged to acquire 12-Lead ECGs on any patient who presents with a new-onset complaint of chest pain, shortness of breath, or altered mental status. Transmit the ECG to the facility the patient will be transported as soon as possible. This is permitted in the TN Rules and Regulations and is allowed for the AEMT scope of practice. Only the paramedics may interpret an ECG. – Updated 2020.

CHEST PAIN (NON-TRAUMATIC)

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of patients with a chief complaint of chest pain.

EMR

- Assess / Stabilize ABC's as needed
- Obtain vital signs

EMT

- Pulse oximetry
- Glucose check
- If BLS ambulance, consider ALS Intercept

AEMT

- Establish IV / Vascular access
- Obtain EtCO₂ reading
- Apply Cardiac Monitor and acquire 12-lead ECG
 - Obtain initial 12-lead on scene and repeat every 10 minutes until arrival at the receiving facility
 - If the patient is less than 25 years old with a normal 12-Lead ECG, then transport without delay and treat as needed
 - If no ST segment elevation is found on the 12-Lead ECG, then consider a 15-Lead ECG and then notify the receiving ED and transmit the ECG.
- Administer 4 chewable 81 mg Aspirin
 - Withhold Aspirin if the Patient is on Anticoagulants
 - Consult on-line medical control if unsure

If a patient is having active chest pain and has a systolic BP > 100mmHg, proceed with giving Nitroglycerin.

- Administer: Nitroglycerin sublingual, 1 tablet or spray, 0.4 mg
- Once IV is obtained, may repeat twice as needed, for a maximum of 3 doses total, to be given 5 minutes apart PRN.
- Ensure that BP remains stable before giving any subsequent doses of NTG.

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CHEST PAIN (NON-TRAUMATIC) (Continued)**CAUTION!**

DO NOT GIVE NTG IF THE PATIENT HAS TAKEN MEDICATIONS USED TO TREAT PULMONARY HYPERTENSION OR ERECTILE DYSFUNCTION, SUCH AS PHOSPHODIESTERASE INHIBITORS (PDEs), EXAMPLE INCLUDE: (DRUGS ENDING IN “FIL”, VIAGRA (SILDENAFIL), CIALIS (TADALAFIL), ETC. IF UNSURE, CONTACT ON-LINE MEDICAL CONTROL.

PARAMEDIC

If chest pain is not relieved after 3 doses of Nitroglycerin, consider Morphine 2 mg IVP.

- Blood pressure must still be above 100 systolic before giving Morphine (or Fentanyl).
- Repeat at 2 mg increments, every 5 minutes, as needed if pain persists, up to a MAX OF 10 MG.
- Fentanyl 50 mcg may be given and repeated once, if Morphine is contraindicated.
 - **If chest pain is due to recent Cocaine usage, consider Versed 2 mg IV or 5 mg IM.**

CHEST PAIN/ACS/STEMI

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of patients with a chief complaint of chest pain.

EMR

- Assess ABCs and Stabilize as needed
- Oxygen and airway maintenance appropriate for the patient's condition
- Obtain vital signs

EMT

- Pulse oximetry
- Glucose check
- If BLS ambulance, consider ALS Intercept
- ECG Monitor
- Oxygen 2-4 LPM via nasal cannula. **DO NOT WITHHOLD**, if the patient is in distress or hypoxic – give 100% via NRB.
- Apply cardiac monitor, acquire 12-Lead ECG ASAP.
 - Any signs of ST-elevation in 2 or more contiguous leads of ≥ 1 mm elevation, send a STEMI alert to the receiving facility.
 - Transmit ECG without delay, transport emergency to nearest appropriate PCI capable facility.

AEMT

- Establish IV / Vascular access
- Obtain EtCO₂ reading
- Administer 4 chewable 81 mg Aspirin, unless the patient is allergic or currently on anticoagulants.
- Give 1 NTG tablet or spray sublingual if systolic blood pressure is greater than 100, every 5 minutes up to a maximum of 3 doses, until pain is relieved or systolic blood pressure falls below 100.
- NTG IS CONTRAINDICATED WITH RIGHT SIDED INVOLVEMENT (INFERIOR MI).

CAUTION!

DO NOT GIVE NTG IF THE PATIENT HAS TAKEN MEDICATIONS USED TO TREAT PULMONARY HYPERTENSION OR ERECTILE DYSFUNCTION, SUCH AS PHOSPHODIESTERASE INHIBITORS (PDEs), EXAMPLE INCLUDE: (DRUGS ENDING IN "FIL", VIAGRA (SILDENAFIL), CIALIS (TADALAFIL), ETC. IF UNSURE, CONTACT ON-LINE MEDICAL CONTROL.

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CHEST PAIN / ACS / STEMI (Continued)

PARAMEDIC

- Any signs of ST-elevation in 2 or more contiguous leads of ≥ 1 mm elevation, send a STEMI alert to the receiving facility.
- Transmit ECG without delay, transport emergency to nearest appropriate PCI capable facility.
- If chest pain is not relieved after 3 doses of Nitroglycerin, consider Morphine 2 mg IVP.
 - Blood pressure must still be above 100 systolic before giving Morphine (or Fentanyl).
 - Repeat at 2 mg increments, every 5 minutes, as needed if pain persists, up to a MAX OF 10 MG.
 - Fentanyl 50 mcg may be given and repeated once, if Morphine is contraindicated.
- Treat any dysrhythmias accordingly.

12-LEAD AND 15-LEAD REFERENCE CHART**Table 1. Infarction Distribution With ST-Segment Elevation Myocardial Infarction And Consequences^{4,15-17}**

ST Elevations	Affected Coronary Artery	Area of Damage	Complications
V ₁ through V ₄	Left coronary artery: Left anterior descending	Anterolateral heart wall Septum Left ventricle His bundle Bundle branches	Left ventricular dysfunction: Decreased carbon dioxide, congestive heart failure Left bundle-branch block Right bundle-branch block Left posterior fascicular block Infranodal block (2° or 3°)
V ₅ through V ₆ , I, aVL	Left coronary artery: Left circumflex branch	Left lateral heart wall	Left ventricular dysfunction: Decreased carbon dioxide, congestive heart failure Infranodal block (2° or 3°)
II, III, aVF, V ₄ R	Right coronary artery: Posterior descending branch	Inferior heart wall Right ventricle	Hypotension (particularly with nitroglycerin and morphine, which can decrease preload) Supranodal 1° heart block Atrial fibrillation/flutter, premature atrial contractions Infranodal block (2° and 3°) Papillary muscle rupture (murmur)
V ₈ and V ₉ (or ST depressions in V ₁ and V ₂)	90% Right coronary artery: Posterior descending branch 10% Left coronary artery: Left circumflex branch (will see elevations in V ₅ through V ₆)	Posterior heart wall	Hypotension Supranodal 1° heart block Infranodal block (2° and 3°) Atrial fibrillation/flutter, premature atrial contractions Papillary muscle rupture (murmur)

15-Lead Myocardial Infarction Reference Chart

How to do:
"15-Lead ECG"



Posterior ECG lead placement

Figure 1 - Posterior ECG lead placement

V7
Left posterior axillary line:
in the same horizontal plane as V4-V6

V8
Tip of the left midscapula:
in the same horizontal plane as V7-V9

V9
Left paraspinal region:
in the same horizontal plane as V4-V6

V1-V3
Should remain unchanged from standard 12-lead ECG

→ V4 Lead

→ V5 Lead

→ V6 Lead

Posterior MI is suggestive by the following changes in V1-V3:

- *Horizontal ST depression
- *Tall, broad R waves (>30ms)
- *Upright T waves
- *Dominant R wave (R/S ratio >1) in V2

Please note that V6 is a good reference point for the horizontal placement of the posterior electrodes V7-9.

If you don't have access to a 15 or 18 lead ECG machine, then leave V1-3 in their normal position and use V4-6, these leads will then become V7-9.

VENTRICULAR FIBRILLATION / PULSELESS V-TACH

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of patients with a chief complaint / signs and symptoms of cardiac arrest in ventricular fibrillation or pulseless ventricular tachycardia.

Per the AHA Guidelines, you no longer have to perform CPR prior to defibrillation. Shock immediately!

AEMT level may give Epinephrine 1:10,000 1 mg IV or IO in cardiac arrest situations under the direction of a paramedic.

- In order to be able to administer this medication, the AEMT must be checked off by their training officer and paperwork, including the TN State EMS form, must be on file if the AEMT received their AEMT licensure prior to January 1, 2023.

EMR

Focus on HIGH QUALITY CPR

- Apply AED / Defibrillator Pads
- Compressions at 100-120 / min At least 2 – 2.4 inches deep.
- Allow adequate chest recoil.
- Minimize interruptions, no more than 10 seconds without compressions.
- Reassess every 2 minutes
- Insert OPA and ventilate with BVM attached to high flow O2

EMT

- Pulse oximetry
- Glucose check
- If BLS ambulance, consider ALS Intercept

AEMT

- Establish IV / Vascular access
- Obtain EtCO2 reading
- Supraglottic airways may be used if unable to intubate due to difficult airway or COVID.
- AEMT's may be able to administer 1 mg Epinephrine 1:10,000 IV or IO under paramedic direction.
 - Repeat every 3-5 minutes as shockable cardiac arrest persists.

PARAMEDIC

- Defibrillate as soon as possible! (**RESUME CPR**)
- Initiate shocks at 120 Joules, 150 Joules, 200 Joules, stay at 200 Joules for other shocks.
 - repeat shocks as needed, no limit on defibrillation attempts.
- Perform rhythm checks (Pulse check) every 5 cycles of CPR

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VENTRICULAR FIBRILLATION / PULSELESS V-TACH (Continued)

- Oropharyngeal intubate the patient ASAP and assess capnography for confirmation of tube placement.
 - Provide 1 breath every 6-8 seconds once a secured airway is in place. (Titrate to effect for a target ETCO₂ upon ROSC).
 - Once a secured airway is in place, chest compressions may be continuous.
- Administer Epinephrine 1 mg 1:10,000 IV/IO
 - Repeat every 3-5 minutes as shockable cardiac arrest persists.
- Administer Amiodarone 300 mg IV/IO Bolus
 - Repeat once in 3-5 minutes at 150 mg IV/IO if arrhythmia persists.
- Or, If Amiodarone is unavailable, use Lidocaine 1-1.5 mg/kg IV/IO bolus.
 - Repeat once in 3-5 min at half initial dose, 0.5-0.75 mg/kg if arrhythmia persists, Max total dose 3 mg/kg.
- Or, If Torsades De Pointes is present, consider Mag Sulfate 2 grams mixed in 100 ml NS Bolus infusion IV/IO.
- Consider H's and T's

BLS CARDIAC ARREST

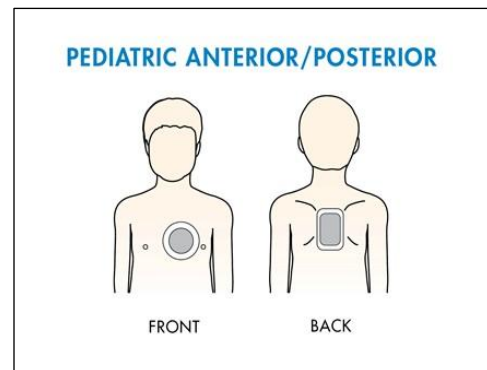
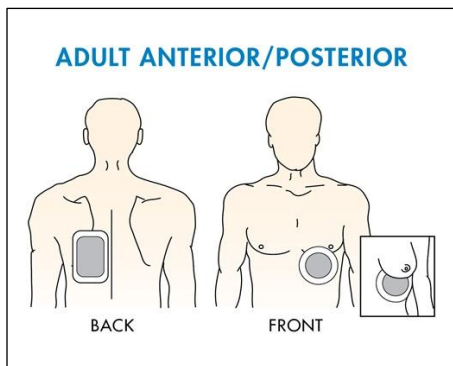
Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of patients with a chief complaint of cardiac arrest.

EMR	EMT	AEMT	PARAMEDIC
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Treatment

- Perform high quality CPR
- A.E.D. arrives at the side of the patient.
- Expose patient's bare chest as modestly as possible
- Wipe patient's chest dry of any water, sweat, or fluids
- Turn on the A.E.D. and attach pads to patient's bare chest



- Pause CPR...Press "Analyze" or continue to follow A.E.D. prompts... stand clear of the patient, do not touch them while the A.E.D. is analyzing.
- **If Shock is Advised**, Be sure that yourself and others are clear of the patient!
 - Not touching the patient.
 - Patient not in standing water or lying on metal. **DELIVER THE SHOCK**
 - **Immediately resume CPR!**
- *Reassess and re-analyze every 2 minutes, and repeat shocks (defibrillations) as long as AED indicates shocks are needed.*
- **If No Shock Advised**
- Resume CPR
- Prepare for ALS personnel arrival.
- Perform actions up to your scope of practice:
 - Vascular Access
 - Supraglottic Airway
- **If a BLS unit, transport immediately if you can be at the ER before an ALS unit can arrive.**

Reference the "Maternal Resuscitation / Cardiac Arrest" Protocol as needed, OB/GYN section.

LUCAS 3 MECHANICAL CPR DEVICEINDICATIONS:

- The LUCAS device is used to perform mechanical chest compressions on patients in cardiac arrest.
- The LUCAS is intended for use on patients' chest depth from 6.7 inches to 11.9 inches with a maximum chest width of 17.7 inches.

CONTRAINDICATIONS:

- If there is no indication that chest compression is likely to help the patient.
- Patients who do not fit within the LUCAS device.
- Patients' chest is too wide or tall to fit in the LUCAS device.
- Patients' chest is too small (pressure pad does not contact the sternum when the sternum pad is in its lowest position).

APPLICATION:

- **Power on the LUCAS device:**
 - Push ON/OFF to start self-test and power up the LUCAS device.
 - The device will be ready and in the ADJUST mode.
 - A second rescuer continues manual CPR.
- **Place the LUCAS BACK PLATE:**
 - Pause manual CPR briefly.
 - Put the BACK PLATE under the patient, immediately below the armpits.
 - A second rescuer needs to assist placing the BACK PLATE and then continuing CPR once the BACK PLATE is placed.
- **Attach the UPPER PART:**
 - Pull the RELEASE RINGS once to open CLAW LOCKS, then let go of the rings.
 - Stop manual CPR briefly while attaching the UPPER PART to the BACK PLATE, and listen for the "CLICK" sound during attachment.
 - Pull up once to assure attachment.
 - A second rescuer continues CPR as long as possible, and helps to attach the UPPER PART when needed.
- **Push down SUCTION CUP. Adjust position if needed.**
 - Push down the SUCTION CUP.
 - The lower edge of the SUCTION CUP should immediately above the end of the sternum.
 - Adjust if necessary (stay in ADJUST mode).
 - A second rescuer can help assist.
- **Lock position. Start compressions.**
 - Push PAUSE to lock START POSITION.
 - Push ACTIVE (continuous) or ACTIVE (30:2) to start compressions.
 - A second rescuer can help assist. - Attach the stabilization strap to prevent downward movement of the LUCAS device during use. - Secure patient's wrist with patient straps on LUCAS device for easier movement of patient. Follow CPR protocols.

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PRECAUTIONS!

Bruising, injury, vomiting, fractured or broken ribs, and soreness of the chest can be possible side effects of performing CPR. Side effects of CPR are common and considered acceptable ramifications given that the alternative is clinical death. If the patient is resuscitated, evaluate and assess for post CPR related injuries.

LUCAS[®] 3

CHEST COMPRESSION SYSTEM

QUICK REFERENCE GUIDE

LUCAS is intended for use as an adjunct to manual CPR when effective manual CPR is not possible (e.g., during patient transport or extended CPR when fatigue may prohibit the delivery of effective/consistent compressions to the victim, or when insufficient EMS personnel are available to provide effective CPR). Refer to the Instructions For Use for complete directions for use, indications, contraindications, warnings, precautions and potential adverse events.

EN



RESCUER 1 (LUCAS Operator)	RESCUER 2
1. POWER ON LUCAS.	
 - Push ON/OFF to start self-test and power up the LUCAS device - The device will be ready and in the ADJUST mode	Provide manual CPR 
2. PLACE THE LUCAS BACK PLATE.	
 - Pause manual CPR briefly - Put the BACK PLATE under the patient, immediately below the armpits	- Assist BACK PLATE placement - Resume manual CPR 
3. ATTACH THE UPPER PART.	
 - Pull the RELEASE RINGS once to open CLAW LOCKS. Then, let go of the rings. - Stop manual CPR briefly while attaching the UPPER PART to the BACK PLATE. Listen for a "CLICK" sound. - Pull up once to ensure attachment	- Continue manual CPR as long as possible - Help to attach the UPPER PART 
4. PUSH THE SUCTION CUP DOWN. ADJUST THE POSITION IF NEEDED.	
 - Push down the SUCTION CUP - The lower edge of SUCTION CUP should be immediately above the end of the sternum - Adjust if necessary (stay in ADJUST mode)	Assist 
5. LOCK POSITION. START COMPRESSIONS.	
 - Push PAUSE to lock START POSITION - Push ACTIVE (continuous) or ACTIVE (30:2) to start compressions	Assist 
... ATTACH STABILISATION STRAP. FOLLOW CPR PROTOCOLS.	

ASYSTOLE / PULSELESS ELECTRICAL ACTIVITY (PEA)

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

AEMT level may give Epinephrine 1:10,000 1 mg IV or IO in cardiac arrest situations under the direction of a paramedic.

- In order to be able to administer this medication, the AEMT must be checked off by their training officer and paperwork, including the TN State EMS form, must be on file if the AEMT received their AEMT licensure prior to January 1, 2023.

The purpose of this protocol is for the monitoring and treatment of patients with a chief complaint / signs and symptoms of cardiac arrest.

- Signs and Symptoms
 - No pulse or respirations
 - Confirm cardiac rhythm with electrodes in 2 leads on cardiac monitor
 - Record in two leads to confirm asystole and to rule out fine V-Fib Presence of electrical cardiac rhythm without a palpable pulse
 - Confirm rhythm with electrodes in two leads
- Consider Underlying Causes and Treat ASAP:

○ Hypoxia	Tamponade
○ Hypovolemia	Thrombosis
○ Hydrogen Ion	Tension Pneumothorax
○ Hypo/Hyperkalemia	Trauma
○ Hypothermia	Toxins

EMR

Focus on HIGH-QUALITY CPR according to AHA's current guidelines!

- Compressions at 100-120/min.
- At least 2 - 2.4 inches deep.
- Allow adequate chest recoil.
- Minimize interruptions, no more than 10 seconds without compressions.
- Apply and utilize AED as soon as it is available.
- Perform rhythm checks every 5 cycles of CPR (every 2 minutes)
- Administer high flow oxygen by BVM with OPA inserted (prn) according to AHA's current guidelines
- Consider Supraglottic Airway device placement
- Follow the Return of Spontaneous Circulation (ROSC) protocol as needed

EMT

- If BLS ambulance, consider ALS intercept
- Transport Emergency

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ASYSTOLE / PEA (Continued)

AEMT

- Establish IV/IO
- Obtain EtCO₂ reading
- Place Supraglottic airway device
- AEMT's may be able to administer 1 mg of Epinephrine 1:10,000 IV or IO under paramedic direction.
 - May repeat every 3-5 minutes.

PARAMEDIC

- Administer Epinephrine 1 mg 1:10,000 IV/IO.
- Repeat every 3-5 minutes.
- Can consider Sodium Bicarb 1 mEq/kg of 8.4% for adults in unwitnessed arrests or witnessed arrests >15 minutes.
- Consider calcium chloride 1 gram IV/O in renal patients.
- Oropharyngeal intubate the patient ASAP and assess capnography for confirmation of tube placement.
- Supraglottic airway may be used if unable to intubate due to difficult airway or COVID.

Special Considerations:

- Reference the "Maternal ACLS/Cardiac Arrest" protocol as needed, in the "OB/GYN" section.
- Waveform kept monography with a reading of less than 10 mmHg may indicate poor CPR or suggest consultation of online medical control to terminate resuscitation efforts
- Persistent asystole despite >20 minutes of resuscitation may I also suggest a consult with online medical control to consider termination of efforts.
- Once a secured advanced airway is in place, chest compressions may be continuous.
- Consider Anterior to Posterior AED/Defibrillation Pad placement

POST-RESUSCITATION (ROSC) MANAGEMENT

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of patients with a chief complaint / signs and symptoms of return of spontaneous circulation.

INDICATIONS: Field ROSC

ACTIONS:

- The Primary Focus is to optimize oxygenation and ventilation.
- Be Careful to ventilate the patient with appropriate rates and volumes
- Ensure head of bed is elevated 30 degrees, if possible.

EMR

- Oxygen and airway maintenance appropriate to the patient's condition including oral suctioning, not to exceed scope of practice.
- Obtain vital signs: Heart rate, pulse, and blood pressure.
- Supportive care

EMT

- Oxygen and airway maintenance appropriate for patient's condition.
- Supportive care.
- Pulse oximetry.
- Blood Glucose check.
- 12 Lead EKG and transmission if applicable.
- If BLS ambulance, consider ALS intercept

AEMT

- Obtain vascular access IV/IO
- Assess BP – If systolic <90 mmHg, administer 250 mL NS or LR bolus (peds systolic BP 70 + (2x age in years), 20 ml/kg bolus) repeat until BP >90 mmHg or appropriate for pediatric age.
- Check the patient's blood glucose level. If low, titrate dextrose 50% prn slowly until normal levels achieved. Try to avoid large swings in serum glucose levels (Peds – See glucose dosing chart on the next page).

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POST-RESUSCITATION (ROSC) MANAGEMENT (Continued)

Glucose (dextrose)	D50 1-2 mL/kg	> 8 years
	D25 2-4 mL/kg	6 months - 8 years
	D10 2-4 mL/kg	neonate - months
	Max Rate 2mL/kg/Min	
If D25 or D10 are not available, utilize a syringe of D50. To make D25, expel 25 mL of D50 and draw up 25 mL of NS. To make D10, expel 40 mL of D50 and draw up 10 mL of NS.		
*Reminder IO is appropriate after 2 failed IV attempts or 90 seconds		

PARAMEDIC

- **If anti-arrhythmic administered:**
 - If the ROSC patient who has received an initial dose of anti-arrhythmic medications goes into cardiac arrest again, proceed to second dose for the antiarrhythmic given...Example: Amiodarone given at 300 mg in V-fib, patients get ROSC, if condition changes and Amiodarone is needed again, it would be at the subsequent dose of 150 mg.
- **Initiate Maintenance Drips for Antiarrhythmic:**
 - If the patient converted with Amiodarone, start an Amiodarone drip at 150 mg and run over 10 minutes.
 - If the patient converted with Lidocaine, start a Lidocaine drip at 1-4 mg/min, reference the formulary as needed for instructions.
- For continued hypotension not responsive to fluid bolus, consider use of Epinephrine push-dose pressors or continuous Epinephrine infusion (reference "Shock Protocol").

Continue ventilator support to maintain $ETCO_2 > 20$.

- ADULTS – Respirations <12 ideally.
- SCHOOL AGE – min resp. rate should be 20-30.
- INFANT – PRESCHOOL – min resp. rate should be 30.

Titrate the artificial ventilator rate to yield an $ETCO_2$ range of 35-45 mmHg.

If patient does not tolerate ET tube or other advanced airway, contact Medical Control for Versed 2-5 mg IV/IO for patient sedation. Reference the "Post Intubation Management" protocol, as needed.

V-TACH WITH PULSE / WIDE COMPLEX TACHYCARDIA

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of patients with a chief complaint / signs and symptoms of symptomatic ventricular tachycardia or wide complex tachycardia with a pulse.

INDICATIONS:

- Confirm and record cardiac rhythm with electrodes in two leads
- Check for palpable carotid pulse
- Decreased / altered mental status
- Dyspnea
- Chest pain / discomfort, suspected AMI
- Hypotension (systolic BP < 90 mmHg)
- CHF / pulmonary edema
- Heart rate > 150 /min and QRS > .12 sec (3 small blocks)

EMR

- Assess ABC's, stabilize as necessary.
- Oxygen and airway maintenance appropriate to the patient's condition
- Obtain vital signs within scope of practice.

EMT

- Obtain and transmit a 12 lead ECG.
- Pulse oximetry
- Obtain blood glucose.
- If BLS unit, consider ALS intercept

AEMT

- Establish IV
- Obtain EtCO2 reading

CONTINUES ON NEXT PAGE

V-TACH WITH PULSE/ WIDE COMPLEX TACHYCARDIA (Continued)

PARAMEDIC

- If in Torsades de Pointes: Administer Magnesium Sulfate 2 grams IV/IO.
- The patient is considered **UNSTABLE** if they are exhibiting one or more of the following:
 - Hypotension
 - Altered Mental Status
 - Acute heart failure
 - Ischemic chest discomfort

If patient is conscious / semi-conscious: give available sedation agent prior to cardioversion / defibrillation. **(Only one sedative may be administered)**
- First choice sedative: Etomidate (Amidate) 10 mg IV/IO.
- If Etomidate is unavailable, then give second choice sedative: Versed 5 mg IV/IO.
- **If no IV/IO is established, consider Versed 5 mg IM.**
- Proceed to synchronized cardioversion/defibrillation.

Monomorphic (regular)
Give synchronized shock (cardioversion) –
• 100 Joules, 150 Joules, 200 Joules

Polymorphic (irregular) / Torsades de Pointes
Give unsynchronized shock (defibrillations) –
• 120 Joules, 150 Joules, 200 Joules PRN

- If patient is **STABLE**:
- Acquire 12 lead ECG to confirm V-Tach prior to treating.
- Ensure the wide-complex tachycardia is regular (consistent R-R intervals)
- Symptomatic, however not yet critical... (treat with medication).
- Consider Amiodarone 150 mg mixed in 100 mL bag of NS infused IV slowly over 10 minutes.
- If Amiodarone is unavailable, Lidocaine 100 mg IV can be given.
- You can repeat at 50 mg in 5-10 minutes.
- Refer to the Amiodarone drip instructions in the formulary section if needed.
- Re-assess and treat appropriately, consult on-line medical control if further orders are needed.
- **TRANSPORT WITHOUT DELAY TO THE NEAREST APPROPRIATE FACILITY.**

PREMATURE VENTRICULAR COMPLEXES (PVCs)

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of patients with a chief complaint / signs and symptoms of a hemodynamic instability due to multiple premature ventricular complexes (PVC's)

EMR

- Basic assessment & management (up to your scope of practice)
- Assess ABC's, stabilize as needed.
- Administer oxygen as needed.

EMT

- Attach pulse oximetry and administer O2 as needed.
- Attach cardiac monitor and acquire 12-Lead ECG. Transmit 12 lead ECG.
- If BLS ambulance, consider ALS Intercept.
- Transport without delay to the nearest appropriate facility.

AEMT

- Establish IV

PARAMEDIC

- Verify if PVCs are present after oxygenation.
- If **NO**, PVCs are not present after oxygenation then Continue to monitor in transport and withhold further treatments unless the patient becomes more symptomatic. Also Consult with on-line medical control if further orders are needed.
- If **YES**, PVCs are present after oxygenation apply treatment if patient exhibits symptoms that suggest hemodynamic compromise as indicated below:
 - Lightheaded / dizziness.
 - Syncope/ near syncope.
 - Ischemic heart disease (severe chest pain).
- Administer Lidocaine 1 mg/kg IV/IO.
- May be repeated at 0.5 mg/kg as needed, up to a maximum of 3 mg/kg.
- Refer to the Lidocaine drip instructions in the formulary section if needed.

If Lidocaine is successful in suppressing ventricular ectopy, begin Lidocaine drip at 2-4 mg/min.
 Infuse desired dose:
 1 mg/min = 15 gtt/min.
 2 mg/min = 30 gtt/min.
 3 mg/min = 45 gtt/min.

Reduced dose by **HALF** if:
 • Patient is over 65 years of age.

SUPRAVENTRICULAR TACHYCARDIA (SVT)

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of patients with a chief complaint / signs and symptoms of symptomatic supraventricular tachycardia.

EMR

- Assess ABCs, stabilize as needed to scope of practice
- Oxygen and airway maintenance appropriate for the patient's condition
- Obtain vital signs

EMT

- If BLS ambulance, consider ALS Intercept
- Obtain 12 lead ECG and transmit
- Pulse oximetry
- Glucose check
- Transport **WITHOUT delay** to the nearest appropriate facility.

AEMT

- Establish IV

PARAMEDIC

- ASAP, obtain 12-Lead ECG, confirm the following before treating as SVT:
 - Is the rate over 150/min in the adult patient?
 - R-R intervals regular? (if not, consider A-fib).
 - QRS width is $\leq 1\text{mm}$ (1 small block).
 - Is there a history suggesting a compensatory tachycardia that just needs fluids.

Stable Patients:

- Attempt vagal maneuvers
- If these are ineffective
 - Begin recording ECG strip...
 - Administer Adenosine:
 - Initial dose - 6 mg rapid IV
 - Immediately follow with a 10 mL NS flush after each dose.

CONTINUED ON NEXT PAGE

SUPRAVENTRICULAR TACHYCARDIA (SVT) – (Continued)

- If no changes are noted then approximately 1 minute later
 - Repeat Adenosine at 12 mg rapid IV immediately followed by 10 mL NS flush.
 - Attempt to record on ECG strip (print).
- If there is still no change following a second dose:
 - Give a third dose of Adenosine, 12 mg immediately followed by 10 mL NS flush.
- The total of all doses should NOT exceed 30 mg.
- Note – If repeat doses of Adenosine do not convert the presumed SVT, it may not be SVT...

Unstable Patient:

- Heart rate over 150/min, **AND** the
- patient exhibits at least one of the following:
 - Hypotension
 - Altered Mental Status
 - Acute Heart Failure
 - Ischemic Chest Discomfort
- If signs and symptoms above are present, then proceed below:
- If the patient is conscious / semi-conscious: give available sedation agent prior to synchronized cardioversion /defibrillation.
 - Only one sedative may be administered.
 - First choice: Etomidate (Amidate) 10 mg IV/IO
 - if unavailable then given 2nd choice: Versed 5 mg IV/IO.
 - If no IV/IO is available, consider Versed 5 mg IM.
- If patient is unconscious then immediately perform synchronized cardioversion.
- Proceed to Synchronized Cardioversion:
 - Place defibrillator in synchronized mode and shock in the following sequence until patient converts:
 - 50 Joules, repeat PRN at 100 Joules.
- Consult with on-line medical control if needed for further orders.
- Note – If repeat doses of Adenosine do not convert the presumed SVT, it may not be SVT...

ATRIAL FIBRILLATION/ATRIAL FLUTTER

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of patients with a chief complaint of atrial fibrillation or flutter.

EMR

- Assess ABCs and Stabilize as needed.
- Oxygen and airway maintenance appropriate for the patient's condition.
- Obtain vital signs.

EMT

- Pulse oximetry.
- Obtain 12 lead ECG and Transmit.
- Glucose check.
- If BLS ambulance, consider ALS Intercept.

AEMT

- Establish IV / Vascular access.
- Obtain EtCO₂ reading.

PARAMEDIC

- Obtain 12 Lead ECG and verify that rhythm is Atrial Fibrillation/Flutter.
- Is patient Stable or Unstable?

STABLE PATIENT

For Atrial Fibrillation or Flutter with a rate greater than 130, the paramedic may consider:

- Lopressor (Metoprolol) 5 mg IVP administered at 1-2 mgs per minute for rate control.
 - Closely monitor heart rate while administering.
- Same dose may be repeated every 5 minutes up to 15 mg's PRN.

CONTINUED ON NEXT PAGE

ATRIAL FIBRILLATION/ATRIAL FLUTTER (Continued)**CAUTION:**

Do not administer if the patient has the following:

- Hypotension (Systolic less than 90 mmHg).
- HR less than 130.
- Congestive Heart Failure (SPO2 less than 92% or rales noted on exam).
- Wide Complex Tachycardia.
- History of Wolf-Parkinson-White (WPW) Syndrome.

UNSTABLE PATIENT

- Classified as altered mental status (outside of their normal mental status) and hypotensive.
- If on-line medical control physician orders to proceed with cardioversion, these are the suggested settings:
 - Atrial Fibrillation – 120 Joules, 150 Joules, 200 Joules.
 - Atrial Flutter – 120 Joules, 150 Joules, 200 Joules.
- Consult On-Line Medical if further orders are needed.
- Transport as indicated.

BRADYCARDIA / AV BLOCKS

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of patients with a chief complaint and signs/symptoms of bradycardia with AV Blocks.

EMR

- Assessment ABCs and Stabilize as needed
- Oxygen and airway maintenance appropriate for the patient's condition
- Obtain vital signs
- Consider H's & T's

EMT

- Pulse oximetry
- Obtain 12 lead ECG and Transmit (NOTE: EMT's and AEMT's can obtain a 12 lead and transmit, but cannot interpret.)
- Glucose check
- If BLS ambulance, consider ALS Intercept
- Transport to nearest appropriate facility

AEMT

- Establish IV / Vascular access
- Obtain EtCO2 reading

PARAMEDIC

- If patient is stable, and time permits, acquire 12-Lead ECG every 10 minutes throughout transport
- Is patient stable or unstable?

Stable:

- Symptomatic, however not yet critical.
- Give Atropine 1 mg IV and repeat every 3-5 minutes PRN with a maximum of 3 mg.
- If Atropine is ineffective or in the presence of a high degree AV block (Mobitz type II or 3rd degree block) consider transcutaneous pacing if the patient's condition warrants.

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BRADYCARDIA / AV BLOCKS (Continued)Unstable:

If Patient exhibits one or more the following signs and symptoms:

- Hypotension.
- Altered Mental Status.
- Acute Heart Failure.
- Ischemic Chest Discomfort.

Then:

- Proceed with Immediate Transcutaneous Pacing:
- (External Cardiac Pacing) – initiated at:
 - 70 PPM (rate) may adjust if needed.
 - 10 mA (energy setting in milliamps).
 - Increase in 10 mA increments until capture.
- If the patient is **responsive** to painful stimulus, or begins to improve and becomes responsive to painful stimulus from pacing, give an available sedating agent:
 - Versed 5 mg IV/IO, repeat PRN to desired effect.
- If no IV/IO is available, consider Versed 5 mg IM (consult on-line medical control for any repeat doses of Versed IM).
- Consider an Epinephrine Drip as an alternative to pacing if needed or while preparing equipment.
 - Infuse Epinephrine Drip 1 mcg/mL solution IV at a rate of 1-4 mcg/min and titrate to desired effect.
 - Reference the Epinephrine Drip instructions in the formulary section if needed.
- Re-assess and treat appropriately, consult on-line medical control if further orders are needed.
- If organophosphate poisoning is suspected to be the cause of the bradycardia, administer Atropine 2 mg IV/IO every 5 minutes until desired effect.

ABDOMINAL PAIN

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of patients with a chief complaint / signs and symptoms of abdominal pain.

EMR**EMT**

- Assess / Stabilize ABC's as needed
- Attach pulse oximetry
- Administer O2 as needed
- Place patient in the position of comfort and transport as indicated

AEMT

- Obtain IV access as indicated: Severe pain, Abnormal vital signs, Needed for administration of analgesia or antiemetic's.
- Treat for shock with LR 500 mL bolus, repeat as necessary and titrate to effect. (Use with caution in CHF and renal failure patients.)

PARAMEDIC

- Paramedics may perform all the noted skills and assessment noted above
- Cardiac monitoring if indicated
- There are no standing orders for analgesics in non-traumatic abdominal pain!

Clinical note: Abdominal pain of moderate to severe acuity should be considered a surgical emergency until proven otherwise and transport should be made without delay in the following:

- **Females or childbearing age / pregnant patients**
- **Patients presenting with S/S indicative of AAA.**
- **Recent post-operative complications**

NAUSEA / VOMITING

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of patients with a chief complaint / signs and symptoms presenting with nausea / vomiting.

AEMT level may give Zofran IV, IM or PO in nausea/vomiting situations under the direction of a paramedic.

- In order to be able to administer this medication, the AEMT must be checked off by their training officer and paperwork, including the TN State EMS form, must be on file if the AEMT received their AEMT licensure prior to January 1, 2023.

EMR

EMT

- Assess / Stabilize ABC's as needed.
- Control airway and be prepared to suction.
- Attach pulse oximetry.
- Give oxygen as indicated.
- Place patient in the position of comfort and transport as indicated.

AEMT

- Obtain IV access, administer fluids as needed for dehydration. 500 mL LR (Normal Saline may be used as an alternative if LR is unavailable) bolus in adults, reassess and repeat as needed to restore normotensive Blood pressure, using caution in patients with renal failure or CHF.
- AEMT's may be able to administer Zofran 4 mg for Adults >40 kg IV, IM or PO under the direction of a paramedic.
 - May repeat once if necessary.

PARAMEDIC

- Paramedics may perform all the noted skills and assessment noted above
- Cardiac Monitor, to rule out underlying acute coronary syndrome, especially in patients > 45 years of age. A patient who has been experiencing emesis may be hypovolemic and have electrolyte depletion as well.
- Acquire and transmit a 12 lead ECG if you suspect any cardiac related illness.
- Adults >40 kg administer Zofran 4 mg IV, IM, or PO (repeat once if necessary).
- Transport as indicated

DEHYDRATION

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of patients with a chief complaint / signs and symptoms presenting with dehydration.

EMR**EMT**

- Assess / Stabilize ABC's as needed.
- Attach pulse oximetry.
- Administer oxygen as indicated.
- Place patient in the position of comfort and transport as indicated.

AEMT

- Obtain IV access: Adults – Administer LR 500 mL bolus, repeat PRN and titrate to effect.
- Use caution in patients with a history of CHF or renal failure.
- NS may be used as an alternative if LR is unavailable
- Use 10 gtt/mL IV tubing to deliver fluids faster

PARAMEDIC

- Paramedics may perform all the noted skills and assessment noted above
- Cardiac monitor should be applied to patients presenting with dehydration due to: Dehydration / hypovolemia can be associated with electrolyte imbalance which may lead to arrhythmias.
- Tachycardia in elderly patients is often associated with hypovolemia.

DIABETIC EMERGENCY / HYPOGLYCEMIA

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of patients with a chief complaint / signs and symptoms of a diabetic emergency / hypoglycemia.

EMR

- Assess / Stabilize ABC's as needed
- Attach pulse oximetry
- Administer O2 as needed
- Place patient in the position of comfort and transport as indicated

EMT

- The use of glucometers to assess Blood Glucose Level (BGL) is within protocol for **EMT-Basics (service trained), AEMT's and Paramedics**. (This is permitted by TN DOH, EMS, Feb. 2017 Clinical Issues Committee Update.
- Determine if the BGL is less than 70 mg/dl, the patient is alert and easily able to maintain their own airway.
- If the patient is awake, alert, and cooperative but is **symptomatic** with a glucose level of 50-70 mg/dl, you may give **Oral Glucose** 15-30 grams (1 tube) PO (oral), buccal, or sublingual.
- *** DO NOT GIVE ANYTHING BY MOUTH TO A PATIENT WHO HAS ALTERED MENTAL STATUS***

AEMT

- If the patient is presenting altered / unresponsive with a BGL < 70, establish IV access. IO access may only be utilized as a secondary only with on-line medical control orders.
- If IV access is successful, line flows well: administer **Dextrose 50% (D50) 25 grams slow IVP**. If D50 is unavailable, Dextrose 10% (D10) may be used as a therapeutic equivalent. D10 containing 10 g / 100 mL should be infused to restore normoglycemia, not to exceed 25 grams (250mL.)
- If dextrose has been given IV, reassess blood glucose level in 5 minutes (not sooner). If glucose level is still less than 70 (refractory hypoglycemia), repeat D50 25 grams slow IVP and reassess again in 5 minutes.
- If unable to secure IV access (or after 3 attempts), and the patient is not awake, alert, and cooperative, give **Glucagon 1 mg IM**.
- Transport as indicated... If a patient's blood glucose and mental status return to normal **prior to transport**, it is acceptable for AEMT's to attempt transport non-emergency.
- Do NOT delay transport to wait on the therapeutic effects of medications
- **If patient is AAOX4 and refusal is obtained, discontinue the IV and make sure the patient eats something with sugar (carbohydrates) prior to leaving, AEMT can obtain the refusal.**

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DIABETIC EMERGENCY / HYPOGLYCEMIA (Continued)

PARAMEDIC

- Paramedics may perform all the noted skills and assessment noted above
- Cardiac monitor as indicated
- Per MD1 if IV access is unsuccessful an EJ is acceptable peripheral access and is preferred over an IO in diabetic patients. IO access may be considered, but requires on-line medical control approval for use in diabetic patients when giving Dextrose.
- If unable to secure IV access (or after 3 attempts), and the patient is not awake, alert, and cooperative, give **Glucagon 1 mg IM**.
- You may repeat Glucagon 1mg IM in 20 minutes if glucose is still less than 70 and LOC is still altered.
- If an IV is established after Glucagon administration, reassess blood glucose level. If glucose is still less than 70 (refractory hypoglycemia), give D50 25 grams slow IVP.
- If the patient is a known alcoholic or otherwise malnourished, administer Thiamine 100 mg IVP, or IM, if no IV access is yet available.

DIABETIC EMERGENCY / HYPERGLYCEMIA

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of patients with a chief complaint / signs and symptoms of hyperglycemia.

EMR

- Assess / Stabilize ABC's as needed
- Attach pulse oximetry
- Administer O2 as needed
- Place patient in the position of comfort and transport as indicated

EMT

- Determine glucose level if >70 and <400, transport as indicated.

AEMT

- Obtain IV access as indicated: if blood glucose is >400, administer IV fluids, 500 mL NS/LR bolus.
- Use caution with CHF and renal failure patients.

PARAMEDIC

- Paramedics may perform all the noted skills and assessment noted above
- Cardiac monitoring as well as ETCO2 with a blood glucose >400

HYPERTENSIVE EMERGENCY

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of patients with a chief complaint / signs and symptoms of a hypertensive emergency.

EMR**EMT**

- Assess / Stabilize ABC's as needed
- Attach pulse oximetry
- Administer O2 as needed
- Place patient in the position of comfort and transport as indicated

AEMT

- Obtain IV access.

PARAMEDIC

- Paramedics may perform all the noted skills and assessment noted above
- Apply cardiac monitor and 12 lead ECG
- If the patient has CVA or AMS symptoms, refer to "Stroke / CVA" protocol and transport as indicated.
- If the patient is pregnant, refer to the "Pre-Eclampsia" protocol.
- If hypertensive blood pressure is secondary to pain, then refer to "Pain Management" protocol.
- If the patient is SYMPTOMATIC with a systolic > 200 or diastolic > 110, and a heart rate greater than 60 bpm:
- Give Labetalol 10 mg slow IVP over 2 minutes
- If there is no change with initial dose, increase dosage to 20 mg slow IVP over 2 minutes every 10 minutes as needed.
- Transport as indicated.

SHOCK PROTOCOL (all types)

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of patients with a chief complaint / signs and symptoms of shock.

EMR**EMT**

- Assess / Stabilize ABC's as needed
- Attach pulse oximetry
- Administer oxygen via NRB
- Place patient in the supine position as tolerated

AEMT

- Obtain IV access (2 large bore is preferred).
- Give fluids 500 mL bolus may be repeated PRN: Lactated Ringers is the preferred fluid.
- Check lung sounds after each bolus

PARAMEDIC

- Paramedics may perform all the noted skills and assessment noted above
- Attach cardiac monitor
- Attempt to determine the etiology of shock by history and exam!
- Hypovolemic-Hemorrhagic shock: Continue with IV fluid bolus as necessary and titrate to effect to maintain a minimum systolic of 90 mmHg. If hypotension is refractory to 30 mL/Kg bolus, consider push-dose Epinephrine if required.
- Anaphylactic shock: Continue with IV fluid bolus and go to Anaphylaxis / Allergic reaction protocol. Push-Dose Epinephrine may be used temporarily until continuous Epinephrine infusion.
- Cardiogenic shock: Go to appropriate protocol. After rate and rhythm normalize and the patient is still in shock, then administer either push-dose Epinephrine as needed or start a continuous Epinephrine infusion at 1-4 mcg/min carefully titrated to effect. Contact on-line medical control for higher dosing.
- Spine shock (neurogenic): Begin Epinephrine Drip at 1-4 mcg/min titrate to effect. Push-Dose Epinephrine may be used as a bridge to continuous infusion. Contact on-line medical control for higher dosing.
- Septic shock: Initiate fluids at 30 mL/kg, move to vasopressors after 30 mL/kg if there is no change. Consider push-dose Epinephrine or continuous Epinephrine infusion at 1-4 mcg/min. Notify the receiving facility of a possible sepsis alert patient.
- TRANSPORT EMERGENCY

SEXUAL ASSAULT

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of patients with a chief complaint / signs and symptoms of sexual assault.

- **Note: in situations where the patient displays acute emotional instability, it is not absolutely required to assess all vitals or other assessments that may require physical contact.**
- **Documentation should provide visual indicators or effective perfusion, and adequate work of breathing as best possible given the situation at hand.**

EMR	EMT	AEMT	PARAMEDIC
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- Assess / Stabilize ABC's as needed.
- Reassure the patient, provide emotional support.
- Treat all injuries.
- Protect the scene and preserve all evidence.
- Ask the patient **NOT** to bathe, change clothes, or go to the bathroom or douche.
- Notify the police, if it has not already been done.
- Place the patient on an open sheet and save the sheet for possible evidence.
- Transport to the hospital with a same gender crew member as patient, if possible. Make efforts to facilitate this by contacting supervisors to request personnel change if necessary.

SICKLE CELL ANEMIA

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of patients with a chief complaint / signs and symptoms of Sickle Cell Anemia.

EMR

EMT

- Assessment / indications supporting need for treatment include: History of sickle cell anemia, signs of infection, hypoxia, dehydration, painful joint(s), and limited movement of joints.
- Assess / Stabilize ABC's as needed
- Obtain vital signs
- Attach pulse oximetry
- Administer O2 as needed
- Supportive care
- Place patient in the position of comfort and transport as indicated

AEMT

- IV access, consider IV bolus at 20 mL/kg (LR)

PARAMEDIC

- Paramedics may perform all the noted skills and assessment noted above
- Cardiac monitoring if indicated (ECG, 12-lead transmit)
- If pain persists / becomes acute: Consult with on-line medical control for order to administer analgesics. **No pain meds are to be given without orders.**
- Attempt to verify history of actual disease process

PSYCHIATRIC EMERGENCIES

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of patients with a chief complaint / signs and symptoms of a psychiatric emergency.

Special note: Ketamine 2-4 mg/kg IM or 1-2 mg/kg IV/IO may be considered, if Versed does not work.

EMR

EMT

- Assure personnel safety and involve law enforcement when needed
- Approach the patient slowly
- Talk in a calm and reassuring tone
- Protect the patient's modesty and promote mutual respect to de-escalate a violent / combative individual
- Assess / Stabilize ABC's as needed
- Administer O2 as needed
- Physical restraint may be needed to facilitate chemical restraint of a dangerous person
- Restrain the patient as needed for patient care and safety
- If restraining and securing is needed, then attempt to restrain the patient to a long spine board, if not then safely restrain them to the cot
- Utilize law enforcement assistance whenever possible

AEMT

- Obtain IV access, if possible, obtain blood glucose level and treat as indicated.

PARAMEDIC

- Paramedics may perform all the noted skills and assessment noted above
- For the acutely agitated and dangerous patient: Consider Versed 2 mg IV or 5 mg IM
- Repeat once and only 10 minutes at 2 mg IV or 5 mg IM
- Further doses **MUST** be approved by on-line medical control
- If chemical restraint is used, once it is safe to do so for the provider, the patient should be assessed with a cardiac monitor, pulse oximetry, and oxygen applied as indicated
- Transport as indicated

RESPIRATORY DISTRESS

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of patients with a chief complaint / signs and symptoms of respiratory distress.

EMT level may give nebulized Albuterol and Atrovent in the presence of wheezing in respiratory distress patients.

- In order to be able to administer these medications, the EMT must be checked off by their training officer and paperwork must be on file.

EMR

- Position of comfort for patient
- Administer O2 if patient is in Respiratory Distress

EMT

- Administer O2 if the patient's O2 saturation is less than 90%
- Assist patient with personal inhaler If wheezing or decreased lung sounds are present or administer Nebulized Albuterol 2.5 mg and Atrovent 0.5 mg, if the patient is wheezing or has decreased lung sounds.
 - Albuterol is contraindicated with a Heart Rate greater than 130 bpm.

AEMT

- Nebulize 2.5 mg Albuterol and 0.5 mg Atrovent if PT is wheezing or has decreased lung sounds
 - Albuterol contraindicated with a Heart Rate greater than 130 bpm.
- Obtain IV access
- Nebulizer treatment may be repeated after 10 min
- Status Asthmaticus – consider administering 0.3 mg Epi 1:1,000 IM once.
 - Caution with Epinephrine administration if the patient is over 60 y/o or cardiac history
- Assess EtCO2

PARAMEDIC

- Consider CPAP/BiPAP
- Obtain 12 Lead EKG
- Consider Mag Sulfate: 2 grams in 100 ml NS infused over 10 minutes.
- Consider Solu-Medrol: 125 mg IV/IM
- Consider intubation/RSI if pt is unable to maintain their airway and/or oxygenate via non-invasive means.

CONTINUED ON NEXT PAGE

RESPIRATORY DISTRESS (Continued)

PARAMEDIC

If Rales is noted, known History of CHF, on diuretics and no recent Fevers

- Wheezing noted then give Albuterol, if not then consider CPAP
- Give Nitroglycerin 0.4 mg SL (one sublingual tablet or spray every 5 minutes to a maximum of 3 doses), if systolic blood pressure $\geq$ 130.
- Consider Morphine 2 mg IV/IO (no repeat dosage).

CONTINUOUS POSITIVE AIRWAY PRESSURE (CPAP)

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of patients with respiratory distress to prevent respiratory failure.

*****CAN BE APPLIED FROM EMT LEVEL THROUGH PARAMEDIC, BUT MONITORED BY PARAMEDIC*****

Indications:

- Obvious signs of moderate to severe respiratory distress
 - i.e. Accessory muscle use or tripod position

Contraindications:

- Respiratory arrest, S/S of Pneumothorax or chest trauma, tracheotomy, active GI bleed or vomiting, unable to follow verbal commands, unable to properly fit CPAP/BiPAP mask, overdoses, and AMS.

EMR

- Assess ABCs and Stabilize as needed
- Oxygen and airway maintenance appropriate for the patient's condition
- Obtain vital signs

EMT

- Pulse oximetry
- Glucose check
- If BLS ambulance, consider ALS Intercept

AEMT

- Establish IV / Vascular access
- Obtain EtCO₂ reading
- Set flow rate (10 lpm) at 6.0-7.0 cmH₂O, prior to placing the mask on the patient's face.

PARAMEDIC

- Attach hose to O₂ source
- Attach in-line EtCO₂ monitor
- Set O₂ flow rate to provide desired pressure settings
- Place mask over the patient's mouth and nose creating an air-tight seal
- Secure mask with provided head strap
- Reassess lung sounds
- Administer Nebulizer treatments as indicated via in-line nebulizer set-up

ALTERED MENTAL STATUS

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of patients with a chief complaint / signs and symptoms of altered mental status.

EMR

- Basic assessment / vitals
 - Underlying causes: head trauma, hypovolemia, hypothermia, hypoglycemia, overdose, etc.
- Oxygen (as needed)
- Consider Narcan 2-4 mg Intranasal (IN), maximum of 8 mg.

EMT

- Assess Glucose
 - <70 refer to hypoglycemia protocol
 - >400 refer to hyperglycemia protocol
- Stroke Assessment

AEMT

- Establish IV Access
- Consider Narcan IN/IM/IV
 - 0.4-2 mg IV or IM, maximum of 8 mg.
 - 2-4 mg IN, maximum of 8 mg.
- Soft Restraints (see "Restraint" protocol)
- Obtain EtCO2 reading if indicated

PARAMEDIC

- Obtain 12 Lead EKG
- Consider Zofran 4 mg IM/IV
- If PT is combative consider: **Versed 2 mg IV or 5 mg IM**
 - Repeat once per protocol and get medical control approval if further is needed.
- If PT is an alcoholic, malnourish or experiencing alcohol withdrawals consider **Thiamine 100 mg IV/IM**

STROKE / CVA

INDICATIONS: AMS (decreased LOC, Coma, Stupor, confusion, seizures, delirium), Headache, Aphasia/Dysphasia, Facial droop, Ataxia, Vision loss/changes.

Transport without delay, ASAP (Emergency) if onset of symptoms is less than 6 hours. If patient is stable and onset is greater than 6 hours non-emergent transport is acceptable.

EMR

- Assessment ABCs and Stabilize as needed
- Establish strong baseline BP with a manual cuff
- Administer O2 2-4 Lpm via NC or 15 Lpm via NRB if pt is hypoxic
- Obtain detailed PT history from pt and family (Onset time, anticoagulants, recent trauma, history of CVA, etc.)
- Perform Stroke scale (Cincinnati, LAMS, MEND – scales included in protocol book for reference)

EMT

- Assess Glucose

AEMT

- Establish IV (prefer 18ga or larger above the hand)
- Obtain EtCO₂ reading

PARAMEDIC

- Obtain 12 lead EKG
- DO NOT TREAT HYPERTENSION without consulting on-line medical control.
- Call stroke alert to receiving facility
- If a Large Vessel Occlusion (LVO) is suspected transport directly to a Comprehensive Stroke Center (Skyline, Vanderbilt, etc.) so long as it does not add more than 15 min to transport time.
 - LVO suspected if two or more of the following occur:
 - Patient states the wrong month and/or age
 - Gaze palsy and/or deviation
 - Arm weakness (hemiplegia or hemiparesis)

THE CINCINNATI PREHOSPITAL STROKE SCALE**Facial Droop** (have patient show teeth or smile):

- Normal: Both sides of face will move equally well.
- Abnormal: One side of face does not move as well as the other side.

Arm Drift (have patient close eyes and hold both arms out):

- Normal: Both arms move the same or both arms do not move at all. Findings, such as pronator grip may be helpful.
- Abnormal: One arm does not move or one arm drifts down compared with the other.

Speech (have the patient say “you can’t teach an old dog new tricks):

- Normal: Patient uses correct words with no slurring.
- Abnormal: Patient slurs word(s), uses inappropriate words, or is unable to speak (For evaluation of acute, non-comatose, non-traumatic neurologic complaint).

Facial/Smile or Grimace:		
Have the patient show teeth or smile.	Normal: Both sides of the face move equally.	Abnormal: Left or right side of face does not move as well.
Arm Drift:		
Have the patient close both eyes and hold both arms straight out for 10 seconds.	Normal: Arms move equally or do not move.	Abnormal: Left or right arm does not move or drifts down.
Speech:		
Have the patient repeat a simple phrase such as “It is sunny outside today”.	Normal: Words stated correctly without slurring.	Abnormal: Patient slurs words or uses the wrong words, or is unable to speak.

MEND PREHOSPITAL STROKE SCALE

MIAMI EMERGENCY NEUROLOGIC DEFICIT (MEND) PREHOSPITAL CHECKLIST					
Date: _____		Name: _____		Age: _____	
Sex: _____					
BASIC DATA			EXAMINATION		
WITNESS NAME: ★		WITNESS PHONE: ★		BP: L _____ / _____ R _____ / _____ Pulse: Rate & Rhythm: _____ Resp _____	
Dispatch time: _____		EMS arrival time: _____		MEND EXAM	
Departure to ED time: _____		ED arrival time: _____			
HISTORY			On scene: Perform LOC & basic exam (Cincinnati Prehospital Stroke Scale in shaded boxes) En route: If time allows, perform the complete MEND exam.		
LAST TIME PATIENT WITHOUT SYMPTOMS ★ DATE: _____ TIME _____					
T-PA EXCLUSIONS		ADDITIONAL HISTORY		MENTAL STATUS	
YES NO ☐ ☐ Head trauma at onset ★ ☐ ☐ Seizure (shaking or staring) at onset ★ ☐ ☐ Taking warfarin (Coumadin) ☐ ☐ History of bleeding problems ☐ ☐ Possible brain hemorrhage (severe headache, stiff neck, ↓LOC)		Symptoms _____ Allergies _____ Medications _____ Past History _____ Last Meal _____ Events Prior _____		CHECK IF ABNORMAL ON SCENE EN ROUTE ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐	
MANAGEMENT			CRANIAL NERVES		
☐ Do NOT treat hypertension ☐ Do NOT allow aspiration → Keep NPO, head up, O ₂ 2-4 L ☐ Do NOT give glucose (unless glucose <50) → IV NS; check fingerstick: _____ ☐ ECG rhythm _____ → If AMI, 12-lead time: _____			B L B L ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐		
STROKE-SPECIFIC ED REPORT (see starred items on checklist)			LIMBS		
SYMPTOM ONSET	NEUROLOGIC EXAM	WITNESS		B L B L ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐	
★ TIME (last time w/o sx/s)	★ Level of consciousness	★ Name			
★ Trauma (history)	★ Speech/language	★ Contact info			
★ Seizure (staring, shaking)	★ Visual fields ★ Moto strength				

LAMS PREHOSPITAL STROKE SCALE

LAMS	Score the Affected Side	Description	Corresponding NIHSS score
Facial Droop	0 Absent	No facial asymmetry. Normal.	Facial Palsy 0-1
	1 Present	Partial or complete lower facial droop	Facial Palsy 2-3
Arm Drift	0 Absent	No drift. Normal.	Motor Arm 0 (Normal)
	1 Drifts down	Drifts down but does not hit the bed within 10 sec.	Motor Arm 1 (drift)
	2 Falls rapidly	Arm cannot be held up against gravity and falls to the bed within 10 sec.	Motor Arm 2-4
Grip Strength	0 Normal	Normal.	No NIHSS for this. Scored 0 if admission neuro exam rated grip strength as 5 (normal)
	1 Weak grip	Weak but some movement.	Admission neuro exam rated grip strength as 4 (weak), 3 (some movement against gravity), or 2 (some movement, but not against gravity)
	2 No Grip	No movement. Muscle contraction be seen but without movement.	Neuro exam rated grip strength as 1 (muscle contraction but no movement) or 0 (no muscle contraction)
TOTAL	= ____ (Range 0-5)		A score ≥ 4 is highly predicted of large artery occlusion

PREHOSPITAL SCREEN FOR THROMBOLYTIC THERAPY

Complete this assessment for all patients symptomatic for Acute Coronary Syndrome or CVA. Report to the Emergency Department Physician/Nurse any positive findings. Document all findings in the EPCR.

Witness/next of kin contact info: _____		
Time of onset of the symptoms: _____		
Systolic BP >240 mmHg	☐ Yes	☐ No
Diastolic BP >110 mmHg	☐ Yes	☐ No
Right arm vs. Left arm Systolic BP difference >15 mmHg	☐ Yes	☐ No
History of recent brain/spinal cord surgery, CVA, or injury	☐ Yes	☐ No
Recent trauma or surgery	☐ Yes	☐ No
Bleeding disorder that causes the patient to bleed excessive	☐ Yes	☐ No
Prolonged CPR (>10 minutes)	☐ Yes	☐ No
Pregnancy	☐ Yes	☐ No
Taking Coumadin, Aspirin, or other blood thinners	☐ Yes	☐ No

INTERFACILITY GROUND TRANSPORT PROTOCOL FOR PATIENTS DURING/AFTER IV tPA ADMINISTRATION FOR ACUTE ISCHEMIC STROKE

- Obtain and record vitals every 15 minutes.
- Obtain and record neurologic checks per the Cincinnati Prehospital Stroke Scale every 15 minutes.
- Follow blood pressure (BP) management per Medication Guide below.
- Patient must be kept strict NPO.
- Maintain head of cot at 30 degrees.

Acute Stroke Management:

• Maintain BP < 180/100. • If BP > 180/100, follow BP protocol below. • If Systolic BP <140 or Diastolic <80 and patient is on antihypertensive drip, titrate down and/or discontinue. • Total tPA infusion time should be 60minutes. • Once tPA infusion completes, hang normal saline with existing tubing to infuse remaining tPA.	*** <u>No other medications are to run through tPA infusion line.</u>*** ***<u>STOP tPa if the patient develops the following symptoms:</u>*** • <u>Worsening LOC.</u> • <u>Severe headache.</u> • <u>Acute hypertension.</u> • <u>Nausea and Vomiting.</u>
---	--

Medication Guide for controlling BP in patients during/after IV tPA administration for Acute Ischemic Stroke:

- If BP > 180/100 and Heart Rate > 60, give Labetalol 10 mg IVP slow x1 over 2 minutes; if no response after 10 minutes, may repeat x1.
- If BP > 180/100 and Heart Rate < 60, ask transferring MD for advice. **DO NOT GIVE LABETALOL!**

Potential Complications:

<u>Symptom(s):</u>	<u>Treatment(s):</u>
Hypotension (Systolic BP < 90):	• Head of bed flat. • Discontinue any antihypertensive drips. • Administer 500 mL normal saline bolus. • If major bleeding suspected, STOP tPA.
Hypertension (BP > 180/100):	• Per medication guide above.
Neurologic Deterioration:	• Assess circulation, airway, breathing (CAB). • Obtain full set of vitals and Neuro check. • Check glucose and treat if < 50.
Airway edema:	• STOP tPA if infusing. • Treat according to Allergic Reaction protocol.
Nausea and Vomiting:	• Treat according to protocol.
Bleeding:	• Apply direct pressure. • Treat according to protocol. • If major bleeding suspected, STOP tPA.

CONTACT SENDING OR RECEIVING FACILITY FOR QUESTIONS

SEIZURES

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of patients with a chief complaint / signs and symptoms of actively seizing with respiratory compromise.

EMR

- Assessment ABCs and Stabilize as needed
- Protect patient from injury
- Suction as needed
- Give O2 and assist ventilations via BVM as needed

EMT

- Assess Glucose

AEMT

- Establish IV
- Obtain EtCO2 reading

PARAMEDIC

- Actively Seizing: Immediately administer Versed 5 mg IM
- Status Seizures: Administer Versed
 - IV – 2 mg (2-3 minutes after the first dose), repeatable once for a total of 3 doses.
 - IM – 5 mg (5 min after first dose), not repeatable, total of 2 doses.
- Special consideration for Pregnant or Post-partum patients (suspected eclamptic seizures)
 - If the patient is greater than 20 weeks pregnant or less than 2 weeks post-partum without a history of seizures then:
 - Mix 4 grams of Mag Sulfate in 100 ml NS and infuse IV/IO over 20 min

*** This can be given in conjunction with Versed ***

SYNCOPE

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of patients with a chief complaint / signs and symptoms of a syncopal episode.

INDICATIONS: Vasovagal, head trauma, hypo/hyperglycemia, hypovolemia, dehydration, shock

EMR

- Assessment ABCs and Stabilize as needed
- Administer O2 as needed

EMT

- Assess Glucose

AEMT

- Establish IV

PARAMEDIC

- Obtain 12 lead EKG
- Special Considerations:
 - Caution on fluid administration with CKD and CHF pts for fluid overload

ANAPHYLAXIS / ALLERGIC REACTION (ADULT)

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of patients with a chief complaint / signs and symptoms of anaphylaxis / allergic reaction.

EMR**EMT**

- Assess / Stabilize ABC's as needed
- Attach pulse oximetry
- Administer oxygen via NRB

AEMT

- Administer Epinephrine (1:1,000): 0.01 mg/kg IM, maximum each dose 0.5 mg. Do not delay Epinephrine, give ASAP.
- Can repeat once after 15 minutes
- Use caution with known cardiac history or age greater than 60 years. If so, consider a maximum of 0.3 mg.
- If wheezing is present: Administer Albuterol 2.5 mg / 3 mL via nebulizer
- Obtain IV access
- If hypotensive or inadequate tissue perfusion, infuse 500 mL bolus of LR or NS, repeat as necessary and titrate to effect.

PARAMEDIC

- Paramedics may perform all the noted skills and assessment noted above
- Apply cardiac monitor
- Administer Antihistamine: H1 Blocker - Diphenhydramine (Benadryl) 25 mg IVP or 50 mg IM
- Administer Steroid: Corticosteroid – Solu-Medrol (Methylprednisolone) 125 mg IV or IM
- If patient is still in extreme Anaphylaxis after treatment above, then initiate continuous Epinephrine infusion at 1-4 mcg/min in adults. Push-dose Epinephrine may be used temporarily until continuous infusion is established.
- Transport as indicated

OVERDOSE-GENERAL / MEDICATIONS

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of patients with a chief complaint / signs and symptoms of an overdose.

**When in doubt, call on-line medical control or TN Poison Control Hotline: 1-(800)-222-1222 or 1-(888)-222-1222
<https://www.vumc.org/poisoncenter/>**

EMR

- Assess / Stabilize ABC's as needed
- Suction as needed
- Oxygen via NRB
- Pulse oximetry
- Aggressive airway control with ventilation if needed
- Obtain history: type and amount of poison, if possible, bring the container with the patient, route of intake, time of intake, history of drug or alcohol usage
- For all personnel, Intranasal (IN) Narcan prefilled nasal spray may be administered 4 mg in one nostril, repeat in 5 minutes as needed. Maximum of 8 mg without further orders.

EMT

- Check blood glucose, if <70 or >400, go to the appropriate hypo/hyperglycemia protocol.

AEMT

- Obtain IV access, any hypotension, then give a fluid bolus of NS 1 liter.
- If a narcotic opiate overdose is suspected: Pinpoint pupils, Hypotension, decreased respirations. Administer Narcan 0.4-4 mg IV or IM, repeat PRN, titrate to effect up to 8 mg. Narcan should be given IM in the absence of vascular access.

PARAMEDIC

- Paramedics may perform all the noted skills and assessment noted above
- Apply cardiac monitor
- Intubate as needed (giving Narcan early in opioid overdose may prevent the need to intubate the patient)
- If beta blocker overdose is suspected and the patient is bradycardic and/or hypotensive: Give Glucagon 1 mg IVP
- If tricyclic overdose is suspected (contact medical control is not sure of the drug): give Sodium Bicarb 8.4% at 1 mEq/kg IV/IO

CONTINUED ON NEXT PAGE

OVERDOSE-GENERAL / MEDICATIONS (Continued)

- If calcium channel blocker overdose is suspected and the patient is bradycardic and/or hypotensive: give calcium chloride 1 gram (10mL) in a 100 mL bag of Normal Saline and given IV/IO over 2-5 minutes
- If the patient is agitated and a possible stimulant overdose is suspected: Consider versed 5 mg IVP or IM. Repeat 2 mg IVP or IM every 5 minutes and titrate to effect. Ketamine can be considered but may only be given with on-line medical control orders
- Transport as indicated

POISONING / CHEMICAL EXPOSURE / HAZ-MAT / NERVE AGENTS

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of patients with a chief complaint / signs and symptoms of poisoning, chemical exposure, Haz-mat, nerve agents.

EMR

- Personnel safety is the highest priority. Do not handle the patient unless they have been decontaminated. All EMS treatment should occur in the "Support Zone" after decontamination
- Assess / Stabilize ABC's as needed
- Aggressive airway control with ventilation if needed
- Suction as needed
- Administer oxygen via NRB
- Obtain history: Type and amount of poison, if possible, bring the container with the patient, route of intake, time of intake, history of drug or alcohol usage
- If inhaled poison, remove patient from the source using appropriate PPE/SCBA preferred. Assess CO levels if exposure to CO is suspected. Consult with / use Haz-Mat personnel when appropriate.
- If the chemical is a dry substance, then brush off the chemical before irrigating, irrigate with copious amounts of water and reassess for hypothermia.

EMT

- Check blood glucose if <70 or >400, go to the appropriate hypo/hyperglycemia protocol

AEMT

- Obtain IV access: Any hypotension, then give a fluid bolus of Normal Saline 1 liter.

PARAMEDIC

- Paramedics may perform all the noted skills and assessment noted above
- Cardiac monitoring as indicated
- Intubate as needed
- If patient is agitated and possible stimulant overdose is suspected: Consider versed 5 mg IV or IM, repeat 2 mg IV or IM every 5 minutes and titrate to effect.
- For organophosphate / nerve agent poisoning: administer Atropine 2 mg IV/IO every 5 minutes until desired effect
- Depending on S/S administer Nerve agent antidote kit: Mild: increased secretions, pinpoint pupils, general weakness: Decontamination, supportive care. Moderate: mild symptoms and respiratory distress, 1 nerve agent antidote kit, may be repeated in 5 minutes, PRN. Severe: Unconscious, convulsions, apnea, 3 nerve agent antidote kits.

ALCOHOL EMERGENCIES

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of patients with a chief complaint / signs and symptoms of an alcohol emergency.

EMR

- Assess / Stabilize ABC's as needed
- Attach pulse oximetry
- Administer O2 as needed

EMT

- Determine blood glucose level, if glucose is less than 70, go to appropriate hypoglycemia protocol.

AEMT

- Obtain IV access as indicated: Significantly altered LOC or unstable vital signs
- If significant Altered Mental Status and possible drug abuse, administer Narcan 0.5 – 4 mg IV or IM. Repeat as necessary or titrate to effect (maximum 4 mg) Narcan may also be administered IM, if IV attempts have been unsuccessful.

PARAMEDIC

- Paramedics may perform all the noted skills and assessment noted above
- Cardiac monitoring if indicated
- Administer Thiamine 100 mg IV or IM if significant Altered Mental Status or malnourished.
- Transport as indicated

SNAKE BITE / ENVENOMATION

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of patients with a chief complaint / signs and symptoms of a snake bite or envenomation.

- The most acute / immediate life threat from a snake bite or other envenomation is anaphylaxis. Immediately proceed to the “Allergic Reaction / Anaphylaxis” protocol as needed.

EMR**EMT**

- Assess / Stabilize ABC's as needed
- Attach pulse oximetry
- Administer O2 as needed
- Splint extremity in a dependent position to restrict movement
- Remove any jewelry from affected extremity
- Keep extremity below the level of the heart
- Transport as indicated, in a supine, resting position to decrease metabolism

AEMT

- Obtain IV access

PARAMEDIC

- Paramedics may perform all the noted skills and assessment noted above
- Cardiac monitoring as indicated
- ETCO2 monitoring as indicated
- If the patient is in severe pain, then see the “Pain Management” protocol.

NEAR DROWNING

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of patients with a chief complaint / signs and symptoms of near drowning.

Transport should be made in all situations involving near drowning. Even in patients who present as stable upon scene, secondary drowning is still a threat to the patient's life. Refusals should involve physician consult via on-line medical control, and absolutely AGAINST MEDICAL ADVICE. Do your best to encourage any patient considering refusal for near drowning to allow transport to a hospital. If possible, assess and document O2 sat and ETCO2 to ensure hypoxia is not present before allowing patient to refuse transport.

If guardians of minors are not allowing transport of a child/minor who had a near drowning event, consult with supervisors, on-line medical control, and involve law enforcement as necessary.

EMR

EMT

- Assess / Stabilize ABC's with attention to C-spine as needed
- Use aggressive airway control and suction as needed
- Attach pulse oximetry
- Administer O2 as needed via BVM or NRB

AEMT

- Obtain IV access

PARAMEDIC

- Paramedics may perform all the noted skills and assessment noted above
- Cardiac monitoring as indicated
- ETCO2 monitoring as indicated
- Consider placement of a gastric tube in patients suffering from submersion injuries / near drowning
- Consider CPAP for fresh water and/or saltwater drowning
- If in cardiac arrest, go to the appropriate protocol
- Transport as indicated

HYPERTHERMIA / HEAT RELATED ILLNESS

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of patients with a chief complaint / signs and symptoms of hyperthermia / heat related illness.

*****Immediate cooling has been proven to be more beneficial if done prior to transport from sporting events. If athletic trainers have cooling capabilities, allow them to cool the patient prior to transport. *****

EMR

EMT

- Assess / Stabilize ABC's as needed
- Assess temperature
- Administer O2 as needed
- If history is suggestive of heat exhaustion or heat stroke: Remove to cooler environment, cool with moist sheets slowly so that the patient will not start to shiver.

AEMT

- Obtain IV access and administer Lactated Ringers (LR) 20 mL/kg

PARAMEDIC

- Paramedics may perform all the noted skills and assessment noted above
- Cardiac monitoring as indicated
- If seizures are present, then go to the seizure protocol.
- Transport as indicated

HYPOTHERMIA

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of patients with a chief complaint / signs and symptoms of hypothermia.

Keep the doors of the ambulance CLOSED with the heat on high, especially in the winter, 3 layers of blankets are recommended for re-warming, with at least one blanket under the patient between them and spine boards, etc.

Note that with hypothermia as subtle as 96° degrees. The body can lose the ability to form clots / slow bleeding in trauma patients.

EMR**EMT**

- Assess / Stabilize ABC's as needed
- Assess temperature
- Remove all wet clothing
- Protect against heat loss and wind chill
- Avoid rough and excessive movement
- Administer oxygen and begin external warming
- If no pulse or breathing: Start CPR. Exception: defibrillate one time and then NO MORE.

AEMT

- Obtain IV access with Normal Saline from fluid warmer: Adult dose, maximum of 1 liter. Pediatric dose: 20 mL/kg
- If narcotic ingestion is possible, give Narcan 0.4-4 mg IV

PARAMEDIC

- Paramedics may perform all the noted skills and assessment noted above
- Cardiac monitoring as indicated
- Monitor cardiac rhythm: treat only life-threatening arrhythmias
- If no pulse or breathing: Resuscitate per ACLS protocol with the following exceptions: Defibrillate 1 time and the NO MORE, Lidocaine may or may not work.
- In pulseless V-tach, V-fib with hypothermia: administer magnesium sulfate 2 grams IVP
- Transport as indicated

ELECTROCUTION / LIGHTNING INJURIES

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of patients with a chief complaint / signs and symptoms of electrocution / lightning injuries.

EMR**EMT**

- Assess / Stabilize ABC's as needed
- Attach pulse oximetry
- Administer oxygen and airway maintenance as needed
- Assessment: Presence of signs and symptoms of electrical injury, Entry/Exit wounds.
- Spinal precautions if electrocution / lightning over 1000 volts or suspicion of spinal injury
- Control gross hemorrhage and dress wounds
- Treat burns per burn protocol

AEMT

- Obtain IV access as indicated

PARAMEDIC

- Paramedics may perform all the noted skills and assessment noted above
- Cardiac monitoring as indicated
- ETCO2 monitoring as indicated
- Reference pain management protocol as needed
- 12 lead ECG, transmit, consult with on-line medical control for treatment of any dysrhythmias.
- Transport without delay
- Consider 2nd IV enroute to hospital

ABDOMINAL / PELVIC TRAUMA

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of patients with a chief complaint of abdominal / pelvic trauma.

EMR

- Assess ABCs and Stabilize as needed
- Oxygen and airway maintenance appropriate for the patient's condition
- Obtain vital signs
- Maintain C-Spine as indicated
- Place patient in position of comfort, with attention to C-Spine needs
- If Pregnant and past the first trimester: Place patient in left lateral recumbent position. Placing pregnant patients in a supine position can compress the vena cava
- Stabilize any penetrating object(s) and place patient supine with legs elevated and flexed at knees and hips
- Cover any eviscerations with saline soaked trauma gauze and place patient supine with legs elevated and flexed at knees and hips

EMT

- Pulse oximetry
- Obtain 12 lead ECG and Transmit
- Glucose check
- If BLS ambulance, consider ALS Intercept

AEMT

- Establish IV / Vascular access
 - If Systolic BP <90 mmHg, infuse LR 20 mL/kg bolus to titrate SBP at permissible hypotension to maintain mental status and distal pulses
- Obtain EtCO₂ reading

PARAMEDIC

- Consider "TXA" Protocol for the unstable trauma patients

AMPUTATIONS

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of patients with a chief complaint of an amputation.

EMR

- Assess ABCs and Stabilize as needed
- Oxygen and airway maintenance appropriate for the patient's condition
- Obtain vital signs
- Stop any life-threatening hemorrhaging
 - Direct Pressure
 - Apply Tourniquet if needed (1 on arm, 2 on leg High and Tight!) Document time of application
 - Consider using hemostatic dressings / gauze
- Place patient in position of comfort
- Care for amputated Part
 - Rinse the amputated part, DO NOT SCRUB!
 - Wrap part in moistened gauze and place in a plastic bag
 - Place sealed bag in a container filled with ice water if available
 - Label container with name, date, and time

EMT

- Pulse oximetry
- Glucose check
- If BLS ambulance, consider ALS Intercept

AEMT

- Establish IV / Vascular access
 - Administer warm Lactated Ringers 500 mL bolus and repeat as necessary and titrate to effect
 - Use caution with patients who have CHF or renal failure
- Obtain EtCO₂ reading

PARAMEDIC

- Consider "TXA" Protocol for the unstable trauma patients
- See "Pain Management" Protocol

AVULSED TEETH

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of patients with a chief complaint of an avulsed tooth/teeth.

EMR

- Assess ABCs and Stabilize as needed
- Oxygen and airway maintenance appropriate for the patient's condition
- Obtain vital signs
- Rinse Avulsed Tooth/Teeth and place in moistened gauze

EMT

- Pulse oximetry
- Glucose check
- If BLS ambulance, consider ALS Intercept

AEMT

- Establish IV / Vascular access
 - If Systolic BP <90 mmHg, infuse LR 20 mL/kg bolus to titrate SBP at permissible hypotension to maintain mental status and distal pulses
- Obtain EtCO2 reading

PARAMEDIC

- Consult On-Line Medical Control regarding re-implantation of avulsed teeth, and proceed as follows if the physician approves:
 - Applicable only for permanent teeth (i.e with patients over 6.5 years of age)
 - Applicable when only one or two teeth are cleanly avulsed and the entire root is present
 - Applicable only to anterior teeth (Front 6, upper, and lower)
 - The patient must be conscious
 - Should be attempted within the first 30 minutes
 - Do NOT force re-implantation, Gentle insertion is all that is necessary. A slightly incorrect position can be corrected later
 - If re-implantation is not feasible and the patient is a fully conscious adult then the best procedure is to place the tooth in the mouth, either under the tongue or in the buccal vestibule (Not recommended for children)

BURNS

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of patients with a chief complaint of burns.

EMR

- Assess ABCs and Stabilize as needed
- Oxygen and airway maintenance appropriate for the patient's condition
- Obtain vital signs
- Stop the Burning!
 - Remove burned or smoldering clothes
 - Remove dry chemicals by brushing off the substance
 - Remove liquid chemicals by flushing with large amounts of water unless contraindicated according to the Emergency Response Guidebook (ERG).
- <10% BSA cover with Cool, Moist, Sterile towels.
- >10% BSA Cover with a dry sterile dressing to prevent heat loss.

EMT

- Pulse oximetry.
- Glucose check.
- If BLS ambulance, consider ALS Intercept.

AEMT

- Establish IV / Vascular access.
 - If Hypotensive, administer LR bolus 500 mL in adults.
- Obtain EtCO2 reading.

PARAMEDIC

- See "Pain Management" Protocol

Burns > 20% BSA that are Partial Thickness or worse, or burns involving the airway, transport to a Burn center: Skyline or Vanderbilt.

CHEST TRAUMA

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of patients with chest trauma.

EMR

- Assess ABCs and Stabilize as needed
- Oxygen and airway maintenance appropriate for the patient's condition
- Obtain vital signs

EMT

- Pulse oximetry
- Glucose check
- If BLS ambulance, consider ALS Intercept
- Place the patient in a position of comfort
- If a flail segment is found, then stabilize it with a bulky dressing
- Open Pneumothorax:
 - Place a VENTED occlusive dressing over the wound
 - If a Tension Pneumothorax develops remove the dressing and let the pressure equal in the chest then replace the dressing

AEMT

- Establish IV / Vascular access
- Obtain EtCO₂ reading

PARAMEDIC

- Needle Decompression and place a 3-way Stop-Cock on the needle to release air as needed
- Intubate and Ventilate as indicated
- See "Pain Management" Protocol

EYE INJURIES

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of patients with signs and symptoms of an eye injury.

EMR

- Assess ABCs and Stabilize as needed
- Oxygen and airway maintenance appropriate for the patient's condition
- Obtain vital signs
- If a chemical injury: Flush with large amounts of sterile water and continue flushing en route
- Treat and cover the eye(s) with a rigid eye patch as needed without placing pressure on the globe

EMT

- Pulse oximetry
- Glucose check
- If BLS ambulance, consider ALS Intercept

AEMT

- Establish IV / Vascular access
- Obtain EtCO₂ reading

PARAMEDIC

- Consult on-line medical control as needed for pain management orders

FRACTURES (GENERAL CARE)

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of patients with signs and symptoms of a potential fracture.

EMR

- Assess ABCs and Stabilize as needed
- Oxygen and airway maintenance appropriate for the patient's condition
- Obtain vital signs

EMT

- Pulse oximetry
- Glucose check
- Immobilize the fracture by securing both fracture ends and the distal and proximal joints
 - Femur Fracture - Apply traction splint as indicated
 - Pelvic Fracture - Stabilize with XP-1, KED, Sheet wrap, padding, etc.
 - Document PMS before, during, and after splinting
- If BLS ambulance, consider ALS Intercept

AEMT

- Establish IV / Vascular access
- Obtain EtCO2 reading

PARAMEDIC

- Consider "TXA" Protocol for unstable trauma patients
- See "Pain Management" Protocol if you do not suspect a head injury and the patient is not hypotensive.

HEAD INJURIES (TRAUMATIC BRAIN INJURY)

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of patients with a potential traumatic brain injury.

EMR

- Maintain C-Spine Precautions
- Assess ABC's
- Obtain vitals (Manual BP, Heart Rate, SpO2)
- Apply oxygen as indicated
- Cardiac Monitoring
- Incline head of spine board or cot 15 degrees

EMT

- Pulse oximetry
- Glucose check
- If BLS ambulance, consider ALS Intercept

AEMT

- Establish IV / Vascular access
- Obtain EtCO2 reading

PARAMEDIC

- Restrain as needed to LSB or soft restraints as indicated for combative patients
- Refer to chemical restrain protocol and consult on-line medical control as needed
- Consider RSI, if required, based on respiratory compromise
- Refer to the pain management protocol as needed
 - If altered due to head injury, consult medical control
 - Document mental status before and after administration of pain medication

PERMISSIVE HYPOTENSION / TRAUMA FLUID RESTRICTION

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of patients with a chief complaint / signs and symptoms of a traumatic injury with intact mental status.

EMR

- Assess ABCs and Stabilize as needed
- Oxygen and airway maintenance appropriate for the patient's condition
- Obtain vital signs

EMT

- Pulse oximetry
- Glucose check
- If BLS ambulance, consider ALS Intercept

AEMT

- Establish IV / Vascular access
 - For Stable Patients (Normotensive and Not Tachycardic):
 - Continuously assess patient for signs of shock and be prepared to administer rapid fluid bolus should the pt begin to show signs of hypotension
 - For Unstable Patients (Tachycardic / Hypotensive / EtCO₂ < 30 mmHg):
 - Administer 20 ml/kg fluid bolus to maintain systolic pressure of 90 mmHg or maintain peripheral pulses
 - Watch EtCO₂ for increase first as initial sign of improvement, then assess for BP increase.
 - Do NOT continue fluids once systolic BP reaches 90 mmHg
 - Repeat fluid bolus as needed to maintain 90 mmHg, Max fluid admin of 60 ml/kg
- Obtain EtCO₂ reading

PARAMEDICConsideration:

- Ensure IV fluids are warm when infusing into trauma patients.

CONTINUED ON NEXT PAGE

PERMISSIVE HYPOTENSION / TRAUMA FLUID RESTRICTION (Continued)

- You may consider use of IO device unless area of injury prohibits IO placement when you are unable to obtain peripheral IVs...fluid administration and rates are the same with IV/IO.

Humeral head IO placement is preferred for a patient that needs fluid resuscitation.

Humeral head IO access is NOT permitted for pediatric patients less than 8 years old, unless authorized by on-line medical control. (updated Jan 2021)

FIELD TRIAGE DECISION SCHEME: THE NATIONAL TRAUMA TRIAGE PROTOCOL

Measure vital signs and level of consciousness

Step One

Glasgow Coma Scale <14 or
Systolic blood pressure <90 mmHg or
Respiratory rate <10 or >29 breaths/minute (<20 in infant < one year)

Yes

Take to a trauma center. Steps 1 and 2 attempt to identify the most seriously injured patients. These patients would be transported preferentially to the highest level of care within the trauma system.

No

Assess anatomy of injury

Step Two

- All penetrating injuries to head, neck, torso, and extremities proximal to elbow and knee
- Flail chest
- Two or more proximal long-bone fractures
- Crush, degloved or mangled extremity
- Amputation proximal to wrist and ankle
- Pelvic fractures
- Open or depressed skull fracture
- Paralysis

Yes

Take to a trauma center. Steps 1 and 2 attempt to identify the most seriously injured patients. These patients would be transported preferentially to the highest level of care within the trauma system.

No

Assess mechanism of injury and evidence of high-energy impact

Step Three

- Falls
 - Adults: > 20 ft. (one story is equal to 10 ft.)
 - Children: > 10 ft. or 2-3 times the height of the child
- High-Risk Auto Crash
 - Intrusion: > 12 in. occupant side; > 18 in. any site
 - Ejection (partial or complete) from automobile
 - Death in same passenger compartment
 - Vehicle telemetry data consistent with high risk of injury
- Auto v. Pedestrian/Bicyclist Thrown, Run Over, or with Significant (>20 mph) Impact
- Motorcycle Crash > 20 mph

Yes

Transport to closest appropriate trauma center which, depending on the trauma system, need not be the highest-level trauma center.

No

Assess special patient or system considerations

Step Four

- Age
 - Older Adults: Risk of injury death increases after age 55
 - Children: Should be triaged preferentially to pediatric-capable trauma centers
- Anticoagulation and Bleeding Disorders
- Burns
 - Without other trauma mechanism: Triage to burn facility
 - With trauma mechanism: Triage to trauma center
- Time sensitive extremity injury
- End-Stage Renal Disease Requiring Dialysis
- Pregnancy > 20 weeks
- EMS Provider Judgement

Yes

Contact medical control and consider transport to trauma center or a specific resource

No

Transport according to protocol

SOFT-TISSUE / CRUSH INJURY

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of patients with a chief complaint / signs and symptoms of a soft tissue or crush injury.

EMR

- Assess ABCs and Stabilize as needed
- Oxygen and airway maintenance appropriate for the patient's condition
- Obtain vital signs
- Cardiac Monitoring
- Place the patient in a position of comfort
- C-Spine protection as indicated
- Stop life-threatening hemorrhaging
- Extremity trauma / exsanguinating hemorrhage- consider tourniquet use
- Apply splints as needed and stabilize penetrating objects
 - Splinting may help prevent secondary injury from dangerously sharp broken bone ends
- Cover open fractures/lacerations and avoid unnecessary movement
- Check PMS before and after moving patient

EMT

- Pulse oximetry
- Glucose check
- If BLS ambulance, consider ALS Intercept

AEMT

- Establish IV / Vascular access
 - If systolic BP < 90 mmHg, infuse NS 20 mL/kg bolus and titrate systolic BP at permissible hypotension or to maintain mental status and distal pulses
- Obtain EtCO₂ reading

PARAMEDIC

- Consider "TXA" Protocol for unstable trauma patients
- See "Pain Management" Protocol

SELECTIVE SPINAL IMMOBILIZATION

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

Spinal immobilization should be performed on the basis of mechanism of injury and the patient's symptoms. If the patient has major mechanism of injury (MOI) or MOI cannot be ruled out, then always immobilize.

EMR	EMT	AEMT	PARAMEDIC
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- Is the patient significantly obtunded? Head injury, drugs, or alcohol. Does the patient have neck / midline spine tenderness? Does the patient have neurologic deficit? If no, immobilization is not necessary.
- If yes, evaluate the mechanism of injury.
- High risk of injury, defined as: High speed MVC, Roll-over MVC, intrusion into the patient compartment, death in the vehicle, fall 2x patient height, fall onto head, auto vs. pedestrian, diving injury, penetrating trauma with motor/sensation deficits.
- Low risk of injury, defined as: low speed MVC, fall from standing position, penetrating trauma without motor/sensation deficits, ambulatory at the scene.
- Pediatric patients low risk injury, no board or collar if: fall from standing, fall from bed, MVA in car seat, neurologically normal for age, no apparent serious injury.
- High risk injury MOI: provide full spinal immobilization
- Low risk injury MOI: apply C-collar, transport in the position of comfort.
- WHEN IN DOUBT, IMMOBILIZE.
- Transport as indicated.
- Special consideration in pregnant patients: If spinal immobilization is required, attempt to tilt the patient 15-30 degrees to her left side in order to keep the baby in vitro from compressing the patient's inferior vena cava causing hypotension.
- If the patient refuses immobilization, all risks are to be explained to the patient and documented in the narrative along with a witness' signature.
- Any deviation from this protocol requires contact with on-line medical control.

SPINAL INJURY

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of patients with a chief complaint / signs and symptoms of a neck/spinal injury.

EMR

- Assess ABCs and Stabilize as needed
- Oxygen and airway maintenance appropriate for the patient's condition
- Obtain vital signs
- C-spine precautions as necessary

EMT

- Pulse oximetry
- Glucose check
- If BLS ambulance, consider ALS Intercept

AEMT

- Establish IV / Vascular access
- Obtain EtCO₂ reading

PARAMEDIC

- See "Pain Management" Protocol, if you do not suspect a head injury and the patient is not hypotensive.

TRAUMATIC CARDIAC ARREST

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of patients with a chief complaint / signs and symptoms of traumatic cardiac arrest.

EMR

- Assess ABCs and Stabilize as needed
- Oxygen and airway maintenance appropriate for the patient's condition
- Begin High Quality CPR
- Obtain vital signs if indicated

EMT

- Pulse oximetry
- Glucose check
- If BLS ambulance, consider ALS Intercept

AEMT

- Establish IV / Vascular access
- Place Supraglottic Airway if Paramedic is not available to intubate
- Obtain EtCO₂ reading
- Consider secondary IV/IO access

PARAMEDIC

- Establish definitive airway most appropriate for patient's condition.
- If Pneumothorax is suspected, reference "Chest Finger Thoracostomy" protocol in the Advanced Procedures section of the protocol
 - Bilateral chest finger thoracostomy may be performed if indicated
 - Consider in blunt or penetrating trauma to the chest
 - Assess for, and treat EARLY into the management of a critical patient
- Consider viability of patient prior to transport
 - Consult with On-Line medical control as needed; refer to "Field Determination of Death" Protocol as needed.
- Transport as indicated with a focus on personnel safety and due regard for others

CONTINUED ON NEXT PAGE

UNCONTROLLABLE EXTREMITY BLEED / EXSANGUINATING HEMORRHAGE

Any state licensed EMR, EMT, AEMT, and Paramedic may perform skills and assessment outline in this protocol for their appropriate level.

The purpose of this protocol is for the monitoring and treatment of patients with a chief complaint / signs and symptoms of an uncontrollable, exsanguinating hemorrhage.

*** BSI, SAFETY!!! GLOVES AND GOGGLES TO PROTECT YOU FROM POTENTIAL EXPOSURE ***

Exsanguinating Hemorrhage?

- Bright red, heavy bleeding from arterial exposure.
- Amputation, de-gloving injuries when acute blood vessel exposure is noted.
- Penetrating trauma with suspected internal hemorrhage evidenced by acute swelling and compensating vital signs.

EXSANGUINATING

EMR	EMT	AEMT
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- Immediately apply the arterial tourniquet (TQ) – CAT or SWAT-T
- For upper extremities, use 1 tourniquet as high on the extremity as possible.
- For lower extremities, use 2 tourniquets, side-by-side as high on the extremity as possible.
- Apply tourniquets until circulation stops and bleeding subsides.
- Consider the use of Hemostatic (Quick Clot) dressing/gauze as needed. If it is a junctional wound then consider packing with hemostatic gauze. **(Hemostatic gauze is located on the RTF kits)**

PARAMEDIC

- Reference the “Pain Management” protocol as needed. Fentanyl may be used. If Fentanyl does not help, then Ketamine can be used in conjunction.

NON-EXSANGUINATING

EMR	EMT	AEMT
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Heavy Dark Red Oozing Bleeding from Suspected Venous Exposure?

- Direct Pressure and elevate the extremity.
- Consider tourniquet application as primary means of hemorrhage control if hemorrhage is significant or refractory to above treatment.
- Consider the use of Hemostatic (Quick Clot) dressing/gauze as needed. If it is a junctional wound then consider packing with hemostatic gauze. **(Hemostatic gauze is located on the RTF kits)**

PARAMEDIC

- Consider saturating gauze with TXA and applying it to the wound.
- Reference the “Pain Management” protocol as needed. Fentanyl may be used. If Fentanyl does not help, then Ketamine can be used in conjunction.

TRANEXAMIC ACID (TXA) ADMINISTRATION – UNSTABLE TRAUMA PATIENTS**PARAMEDIC****INDICATIONS:**

- **Hypotension**, and other signs of shock, associated with known or suspected blood loss.
- **Less than 3 hours** since time of injury.

CONTRAINDICATIONS:

- More critical interventions need to be done for your patient (Do not delay to give TXA).
- Isolated head injury.
- Longer than 3 hours since time of injury.

Adult Dose (16 years and older):

Add 2 gram to 100 mL NS (or D5W) and infuse IV over 10 minutes.

ALL PATIENTS RECEIVING TXA SHOULD BE TRANSPORTED TO A FACILITY THAT IS CAPABLE OF MAINTAINING THE TXA INFUSION.

Both LifeFlight and AirEvac now have TXA that can be given in flight if

TRANEXAMIC ACID (TXA) ADMINISTRATION – STABLE COMPLICATED BLEEDING**INDICATIONS:**

- Controlling Non-Arterial bleeding that is difficult to control with direct pressure, but not needing a tourniquet. (Example: Dialysis Fistula, etc.)
- Uncontrolled Epistaxis (Nose Bleed).

Application to a Wound

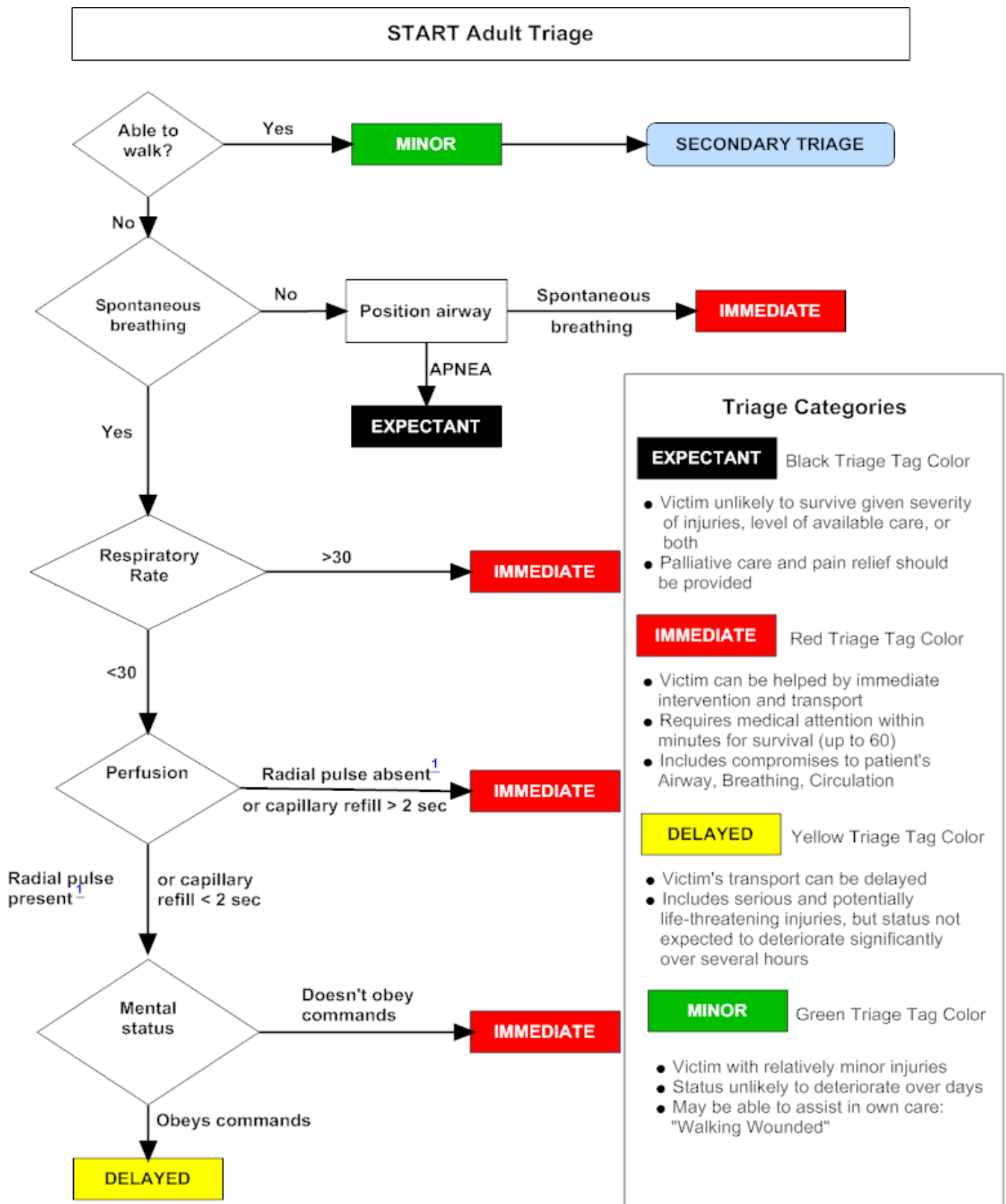
- Draw TXA from the vial, and saturate a sterile gauze pad with it.
- Apply directly over the wound with direct pressure.

Application for Epistaxis

- Administer 0.5-1 mL into the nostril by atomizer.
- Continue pinching the nose together on the soft tissue just below the bridge of the nose.

Application to Post-Dental Procedure

- Draw TXA from the vial, and saturate a sterile gauze pad with it.
- Apply directly over the wound.
- Be sure to instruct the patient to NOT swallow the gauze pad.

START ADULT TRIAGE

OB/GYN NON-DELIVERY COMPLAINTS AND COMPLICATIONS

Any state licensed EMR, EMT, AEMT, Paramedic, may perform skills and assessment outlined in this protocol for their appropriate level.

Assessment questions to consider.

- How many pregnancies (Gravida) & how many births (Para) has the pt had?
- How many weeks (trimester) along is the pt? Estimated Date of Delivery? Multiple births expected?
- Any complications? Hypertension (Pre-Eclampsia), Gestational Diabetes, lower extremity swelling, previous ER visits / admits (current and past pregnancies) any previous birth complications?
- Any vaginal bleeding or discharge? How much? How long has it been going on?
- Any pre-natal care, OB visits, what clinic?
- Any trauma prior to onset of symptoms / complaints?

Placental Abruption (Abruptio Placentae)

- **Vaginal Bleeding:**
 - **Nature:** Often **dark red** and may be concealed (hidden hemorrhage) beneath the placenta.
 - **Pain Association:** **Typically, painful.** The bleeding is usually accompanied by abdominal pain.
- **Uterine Symptoms:**
 - **Uterine Tenderness:** The uterus may feel **firm, rigid, or "board-like"** due to muscle contractions.
 - **Contractions:** Frequent and painful uterine contractions are common.
- **Fetal Distress:**
 - **Possible:** Due to compromised placental function, signs of fetal distress (e.g., abnormal fetal heart rate) may be present.
- **Associated Risk Factors:**
 - **Hypertension:** Chronic or pregnancy-induced hypertension.
 - **Trauma:** Abdominal injury or trauma.
 - **Substance Use:** Particularly cocaine use.
 - **Previous Abruption:** History of prior placental abruption

Placenta Previa

- **Vaginal Bleeding:**
 - **Nature:** **Painless and bright red** bleeding.
 - **Timing:** Typically occurs in the **second or third trimester**, often without any abdominal pain.
- **Uterine Symptoms:**
 - **Uterine Tone:** The uterus is usually **soft and non-tender**, without the rigidity seen in abruption.
 - **Contractions:** Generally **absent** unless there is another underlying issue.
- **Fetal Condition:**
 - **Less Immediate Fetal Distress:** Unless bleeding is severe, fetal distress is less common compared to abruption.

CONTINUED ON NEXT PAGE

OB/GYN NON-DELIVERY COMPLAINTS AND COMPLICATIONS (Continued)

Any state licensed EMR, EMT, AEMT, Paramedic, may perform skills and assessment outlined in this protocol for their appropriate level.

AEMT level may give Zofran IV, IM or PO in nausea/vomiting situations under the direction of a paramedic.

- In order to be able to administer this medication, the AEMT must be checked off by their training officer and paperwork, including the TN State EMS form, must be on file if the AEMT received their AEMT licensure prior to January 1, 2023.

EMR

EMT

- Place in position of comfort. (**3rd trimester: If supine, place in left lateral recumbent. If spinal precautions are used, tilt board, and pad voids.**)
- Look, or have pt feel for protruding limbs, placenta, fluids.
- Control bleeding as appropriate. Do not insert bandaging into vagina.
- Obtain Vitals including blood pressure, heart rate, breathing rate, oxygen saturation.
- Apply oxygen and airway control as appropriate for condition and your licensure level.

AEMT

- If required, initiate IV saline lock or normal saline / LR @ TKO. If signs of shock are present, then 20 ml/kg fluid bolus. May also see “**Shock**” protocol.
- Consider Glucose check.
- If nauseous, may administer Zofran IV or ODT as appropriate. See “Zofran” in formulary for contraindications, considerations, and dosing.

PARAMEDIC

- 3-lead monitoring and obtain 12-lead as appropriate.
 - Pregnancy increases fluid volume and strain on the heart. Always consider cardiac monitoring.

Remember to treat the mother’s complaints and complications. Always call for orders if unsure.

ACTIVE OR IMMINENT DELIVERY

Any state licensed EMR, EMT, AEMT, Paramedic, may perform skills and assessment outlined in this protocol at or below their appropriate level.

Assessment Considerations

- **Term of pregnancy** in weeks, and **original** estimated delivery date?
- Obtain history including number of pregnancies (Gravida) and previous births (Para).
- Any complications with current or previous pregnancy / birth?
- Any pre-natal care or problems?
- Any lower extremity swelling?
- Any other pertinent information?

Important Notes

- The greatest risk to the newborn is airway obstruction and hypothermia. Keep newborn dry, warm, and airway open (if needed; bulb suction mouth then nose, squeeze bulb before insertion.) Always assume respiratory and use positive pressure ventilation if needed.
- The greatest immediate risk to mother is post-partum hemorrhage. Watch for excessive blood loss and hypovolemic shock.
- Consider 2nd unit if either mother or newborn are critical, otherwise keep them together.

EMR

EMT

- Prepare ambulance with OB kit and keep patient area warm. Prepare mother by removing all lower clothing and preserving dignity as much as possible (gown, blanket, or sheet).
- **Monitor vitals including blood pressure, heart rate, and pulse ox. Apply oxygen and airway support as appropriate.**
- Visually check for crowning, limbs (breech), or placenta, repeat as needed. Do not perform internal examination.
- Do not delay transport. Alert receiving facility as early as possible. Transport as indicated. Be aware of signs of imminent delivery, i.e. contractions less than or around 3-5 minutes apart and 30-60 seconds long.
- If head is crowning, prepare to catch, apply gently pressure to prevent an “explosive” delivery. Do not apply force, push / pull newborn.
- As head delivers, suction mouth then nose with bulb suction. Check for nuchal cord around neck, remove or relieve cord gently.
- After delivery, keep infant on same level as mother, **Dry, Warm, Stimulate** infant. If baby is crying, has good muscle tone, heart rate above 100 bpm, and breathing spontaneously place on mother’s chest. **Keep infant warm and covered.** Check APGAR score @ 1 and 5 minutes apart.
- Clamp cord if no longer pulsing or if infant requires resuscitation. Clamp cord at roughly 8 and 10 inches from baby. Cut in between clamps. Re-assess for bleeding and apply additional clamps as needed.
- Re-evaluate for bleeding, signs of shock, additional births, placental delivery. Maintain ABCs for both.

CONTINUED ON NEXT PAGE

ACTIVE OR IMMINENT DELIVERY (Continued)

AEMT

- Initiate an IV (on mother) with LR @ TKO. Increase rate to 20 ml/kg if signs of shock, maintain blood pressure.
- Check BGL heel stick on newborn. Normal BGL at or above 50.

PARAMEDIC

- Apply 3-lead monitoring to mother.
- **Refer to “Neonate Resuscitation” Protocol as needed.**

ABNORMAL DELIVERY: Breech or Limb Presentation

Any state licensed EMR, EMT, AEMT, Paramedic, may perform skills and assessment outlined in this protocol for their appropriate level.

EMR

EMT

- **Breech Presentation**

- Provide oxygen and airway maintenance as appropriate.
- Allow delivery to proceed spontaneously – Do Not Pull.
- Support infant's body as it delivers.
- If head delivers spontaneously then proceed with "Normal Delivery" protocol.
- If head **does not** deliver within **3 minutes**, insert gloved hand into vagina to create an airway for the infant.
- **DO NOT** remove your hand until relieved by Higher Medical Authority.

- **Limb Presentation**

- Position mother supine, with head lowered and hips elevated.
- Transport immediately, do not delay.

AEMT

- Initiate IV TKO with LR or Normal Saline.
- If hypotensive 20 ml/kg of LR or Normal Saline.

PARAMEDIC

- Monitor for arrhythmia's
- Contact Medical Control and Transport Emergently

ABNORMAL DELIVERY COMPLICATIONS:Meconium Staining / Prolapsed Cord

Any state licensed EMR, EMT, AEMT, Paramedic, may perform skills and assessment outlined in this protocol for their appropriate level.

Meconium Staining

EMR	EMT	AEMT
-----	-----	------

- Do not stimulate breathing till you have bulb suctioned **mouth** then **nose**.
- Administer blow-by oxygen as appropriate for patient's condition.
- Obtain APGAR scores.
- Reassess and Repeat treatments.
- Apply Pulse Ox if patient is not improving.

PARAMEDIC

- Apply monitor if not improving.
- Use meconium aspirator if thick secretions causing airway obstruction still present.

Prolapsed Cord

EMR	EMT
-----	-----

- Position mother supine, with hips elevated as high as possible. Use pillows, sheets, blankets, etc. to pad under her hips. Goal is to use gravity to help hold the head off of the cord.
- Instruct mother to "pant" during contractions to help reduce force of contractions.
- Check for pulse in cord.
 - **No pulse in cord:** Insert gloved hand into vaginal, gently apply pressure to infant's head to restore pulse in cord. Cannot move hand after pulse is restored for any reason until instructed by a doctor. Place moist sterile dressing over exposed cord. Transport immediately.
 - **Pulse found in cord:** Cover exposed cord with sterile moist dressing, transport immediately. Reassess regularly to ensure pulse remains intact.

AEMT

- May initiate IV @ TKO if appropriate.

PARAMEDIC

- Monitor for arrhythmias.

ABNORMAL DELIVERY: AMNIOTIC SAC PRESENTATION & NUCHAL CORD

Any state licensed EMR, EMT, AEMT, Paramedic, may perform skills and assessment outlined in this protocol for their appropriate level.

AMNIOTIC SAC PRESENTATION**EMR****EMT**

- Oxygen and airway maintenance appropriate to the patient's condition.
- Place patient in position of comfort.
- Pulse oximetry.
- Amniotic sac:
 - If no fetus visible, cover presenting part with moist, sterile dressing.
- Contact Medical Control ASAP.

AEMT

- Initiate IV NS / LR @ TKO. If hypotensive 20 ml/kg.

PARAMEDIC

- Monitor for arrhythmias

NUCHAL CORD (cord around neck)**EMR****EMT****AEMT****PARAMEDIC**

- Carefully remove cord from around baby's neck by slipping cord over the baby's head if possible.
- Attempt to prevent creating a knot in the cord.
- Prevent cord from strangulating the neonate during delivery.
- Transport as indicated, do not delay.

APGAR Score Reference Chart

APGAR Test Scoring			
	Score 0	Score 1	Score 2
A pppearance			
	Blue all over	Blue only at extremities	No blue coloration
P ulse	No pulse	<100 beats/min.	>100 beats/min.
G rimace			
	No response to stimulation	Grimace or feeble cry when stimulated	Sneezing, coughing, or pulling away when stimulated
A ctivity			
	No movement	Some movement	Active movement
R espiration	No breathing	Weak, slow, or irregular breathing	Strong cry

Score of 0-3 = Severely depressed, RESUSCITATE!!!

Score of 4-7 = Moderately depressed, STABILIZE

ASSESS A.P.G.A.R. SCORES AT 1 MINUTE AND 5 MINUTES POST PARTUM!

NEONATAL RESUSCITATION

Any state licensed EMR, EMT, AEMT, Paramedic, may perform skills and assessment outlined in this protocol for their appropriate level.

EMR

EMT

- Vigorously dry, warm, and stimulate. Clamp and cut cord per delivery protocol. APGAR Assess for tone, respirations, and pulse.
- If respirations are slow, apneic, labored, or persistent central cyanosis.
 - Reposition airway, suction the mouth then nose if secretions are present.
 - Positive pressure ventilation, (BVM no oxygen) for 1 minute.
- Apply pulse ox. Reassess.
- If heart rate less than 100 bpm, do positive pressure ventilation (BVM / Neo-T). Use as little O₂ as possible. Reassess and continue unless HR drops less than 60 bpm or discontinue if pt improves.
- If heart rate less than 60 bpm, start CPR @ 3 compressions to 1 breath with 100% O₂. Reassess, continue if HR stays less than 60 bpm.
- Do not delay transport.

AEMT

- Initiate IV/IO access, bolus NS/LR @ 10 ml/kg if required.
- Obtain blood glucose via heel stick. If less than 50 mg/dl administer D10 @ 2-4 ml/kg bolus and reassess, repeat as needed. Target of 70-150 mg/dl.

PARAMEDIC

- If heart rate less than 100 bpm and not improving with PPV, **consider** advanced airway.
- Place cardiac monitor.
- If heart rate less than 60 bpm **initiate** advance airway.
- If heart rate remains less than 60 bpm despite 3-5 minutes of CPR @ 3:1, then initiate Epinephrine 1:10,000 at 0.02 mg/kg, every 3-5 minutes, till heart rate greater than 60 bpm.
 - Refer to pediatric weight tape to confirm dosages.
- Consider Hypovolemia, Pneumothorax, Hypothermia, Toxins.
- Do not delay transport

PRE-ECLAMPSIA / ECLAMPSIA

Any state licensed EMR, EMT, AEMT, Paramedic, may perform skills and assessment outlined in this protocol for their appropriate level.

Signs and Symptoms

- Headache
- Double vision or seeing spots
- Swelling of face, arms, or legs
- Blood Pressure greater than 140/90
- Seizures
- May occur weeks after delivery

EMR**EMT**

- Assess ABC's.
- Place left lateral when after 20 weeks.
- Administer oxygen therapy as needed.

AEMT

- Initiate IV / IO, may consider a 2nd line.
- Obtain blood glucose.

PARAMEDIC**If symptomatic and systolic greater than 140 or diastolic greater than 90**

- Obtain 12-lead and monitor for arrhythmia's.
- Administer 4 grams of Magnesium Sulfate in a 100 mL bag of NS, over 20 minutes.
- If seizure activity is noted, may give Versed 2 mg IVP or 5 mg IM.
- If still hypertensive after Magnesium infusion, may give 10 mg of Labetalol slow IVP, repeat at 20 mg every 10 minutes till target BP drops below 140/90 or other contraindication of Labetalol occurs.
- Contact Medical Control.
- Do not delay transport, transport as indicated.

MATERNAL ACLS / CARDIAC ARREST IN PREGNANCY**Management of the Unstable Pregnant Patient:**

- The patient should be placed in a full left lateral decubitus position to relieve aortocaval compression.
- Intravenous access should be established above the diaphragm to ensure that the intravenously administered therapy is not obstructed by the gravid uterus.
- Precipitating factors should be investigated and treated.
- Pregnant patients are at an **increased risk of aspiration**, hypoxemia and pulmonary edema.
- Systemic hypotension can overwhelm compensatory mechanisms, which attempt to maintain uterine blood flow.

**Chest Compressions in Pregnancy:**

- Guidelines for hand placement, rates, and ratios are the same as with non-pregnant adult patients.
- Keep supine, implement manual LUD during compressions, starting at ≈ 12 to 14 weeks of gestational age.
- There is no literature examining the use of mechanical CPR devices in pregnancy, this isn't advised at this time.

Transport / Operations:

- Transport to a facility that can perform a cesarean delivery may be required when indicated (For out-of-hospital cardiac arrest or cardiac arrest that occurs in a hospital not capable of cesarean delivery).
- If available, transport should be directed toward a facility that can perform PMCD, but transport should not be prolonged by >10 minutes to reach an ER with more capabilities. (PMCD = perimortem cesarean delivery).

Electrical Therapy (Defibrillation) Issues During Pregnancy:

- Current defibrillation protocols should be used in the pregnant patient as in the non-pregnant patient.
- The lateral defibrillator pad/paddle should be placed under the breast tissue when possible.

Breathing and Airway Management in Pregnancy:

- Oxygen reserves are lower and metabolic demands are higher; earlier ventilatory support may be necessary.
- Intubation starting with an ETT with a 6.0- to 7.0-mm inner diameter is recommended.
- Upper airway edema and friability (weakening of tissue) occur as a result of hormonal effects and may reduce visualization during laryngoscopy and increase the risk of bleeding.
- Renal blood flow increases, stressing the kidneys, increasing the tendency for pulmonary edema to develop.

Emergency (ACLS) Medications in Pregnancy:

- In the setting of cardiac arrest, no medication should be withheld because of concerns about fetal teratogenicity.
- Drug metabolism is altered in pregnancy. However, there are no current recommendations for altering dosages of routine ACLS drugs. **Use standard drugs and doses when indicated.**

FIELD DETERMINATION OF DEATH

EMR	EMT	AEMT
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Resuscitation is not to be attempted or continued if a patient appears to be deceased and having the following signs:

- Rigor Mortis.
- Dependent Lividity.
- Decomposition of body tissue.
- Devastating, non-survivable injury(s) clearly incompatible with life:
 - Decapitation.
 - Incineration.
 - Brain matter visible.

RESUSCITATION MAY NOT BE ATTEMPTED OR CONTINUED IF THE FOLLOWING CONDITIONS ARE:

- A valid DNR order, advanced directive, P.O.S.T., P.O.L.S.T., P.O.A., or patient advocacy paperwork is present or produced and the patient is in full cardiac arrest.
 - If the family states that the patient has a DNR, but they cannot produce the paperwork, then On-Line Medical Control needs to be contacted to get an order to stop CPR.
- An on-scene physician with appropriate identification or Medical Control issues an order to stop CPR.
- If family member(s) / bystander(s) want resuscitative measures started, then begin and transport.

PARAMEDIC

- Blunt traumatic cardiac arrest and the patient is in Asystole.
- Unwitnessed arrest with unknown amount of down time asystole in 2 or more leads.
- The paramedic may choose not to perform an ECG if obvious death is noted because attempts need to be made to preserve any potential crime scene evidence.

WHEN IN DOUBT, START CPR!!!!

NOTE:

CPR prior to crew arrival can be stopped by the crew if the death is obvious. However, if the crew begins CPR, then it can only be stopped with direction from Medical Control. You should feel free to start or continue CPR when in doubt or if family insists.

SUPRAGLOTTIC AIRWAY (iGel) PROTOCOL

ADULT AND PEDIATRIC

EMR	EMT	AEMT	PARAMEDIC
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Any licensed EMR, EMT-B, AEMT, Paramedic, or CCEMTP that operates under Sumner County EMS protocols and standing orders as an employee of EMS or a responder with an affiliated first responder agency may perform this skill if they have been properly trained, credentialed and documented.

Device Models:

Current inventory contains the iGel. A soft-tip suction catheter or OG tube can be inserted here to provide gastric suctioning / decompression.

Latex-Free Device:

The iGel is 100% latex free and should be considered safe to use on patients with latex sensitivity.

INDICATIONS:

- Adult or pediatric patients in need of prolonged ventilatory support that are not improving with 100% O₂ via BVM.
- Absence of a gag reflex.
- To be used as an alternate airway device in the absence of a paramedic or the inability to perform intubation.
- Can be used in medical or trauma patients.
- The iGel airway should be used as a first-line airway for adults with suspected COVID.

CONTRAINDICATIONS:

- Do not use the iGel airway to initially secure the airway for purposes of RSI.
- Do not use in responsive patients with an in-tact gag reflex.
- Do not use in patients who have ingested caustic substances.
 - Acids, Alkalis, Petroleum products.
- Do not use if the patient has a history of liver cirrhosis or esophageal varices.
- Do not use in patients with trismus, limited mouth opening, airway mass or abscess.

PRECAUTIONS:

- The iGel airway is not proven to protect the airway from the effects of regurgitation and aspiration. Orotracheal intubation by a paramedic remains the standard of care for all patients, adult or pediatric.
- Be sure to select the most appropriately sized iGel airway for the specific patient being treated. Reference the chart on this protocol to help choose the best size for each patient.

CONTINUED ON NEXT PAGE

SUPRAGLOTTIC AIRWAY (iGel) PROTOCOL (continued)

ADULT AND PEDIATRIC

Directions for Insertion of the iGel Airway:

1. Stabilize ABC's as needed, Hyperoxygenate the patient with a BVM attached to high flow O₂, insert oral airway if appropriate, and suction as needed.
2. Prepare Equipment: Select the Appropriately Sized iGel Airway.

Size	Patient Criteria (Weight / Height)
1	Newborn (2-5 kg)
1.5	5-12 kg
2	10-25 kg
2.5	25-35 kg
3	30-60 kg
4	50-90 kg
5	90 + kg

Pediatric Color-Coded / Length Based Measurement Tape Identification for iGel Airways

Gray Size 1	Pink Size 1.5	Red Size 1.5	Purple Size 2	Yellow Size 2	White Size 2	Blue Size 2	Orange Size 2 or 2.5	Green Size 2.5 or 3
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3. Apply a water-based lubricant (KY jelly, etc.) to the beveled distal tip and posterior aspect of the tube. Be careful to not introduce lubricant in or near the ventilatory openings.
4. Open airway with appropriate maneuver, place patient's head in a sniffing position.
 - a. Use caution not to hyperextend the neck in patients with suspected C-spine injuries.
 - b. Pediatric patients with a proportionally larger head may need padding under the shoulder blades to help maintain positioning.
5. Insert the airway downward toward the feet. Without exerting excessive force, advance the airway until resistance is met and the integral bite block aligns with teeth or gums.
6. Confirm placement of the device using the following methods:
 - a. Colorimetric device (should turn yellow if getting exhaled CO₂ = good placement).
 - b. Bilateral chest rise and fall (condensation may be seen in the airway tube).
 - c. No epigastric sounds or gastric distention noted upon giving breaths.
 - d. Positive bilateral breath sounds heard upon giving breaths.
 - e. **Waveform capnography (AEMTs and Paramedics can perform) should be used as soon as possible to verify and monitor placement. AEMTs can only monitor the value of capnometry not the capnography waveform.**
7. Secure the device (ETT holder works well) and continue care / transport without delay.

Suction Catheter Sizes per Device

iGel Size	Catheter(s) Size
1.5	10 Fr
2	10-12 Fr
2.5	10-12 Fr
3	10-12 Fr
4	10-12 Fr
5	14 Fr

INTERFACILITY TRANSPORTATION OF BLOOD PRODUCTS

PARAMEDIC

PURPOSE:

Patients may require transportation to another, Medical/Trauma Center with blood or blood products infusing. A paramedic or registered nurse must accompany these patients in the patient compartment, or physician trained in these procedures. The paramedic shall be knowledgeable in the administration of blood, blood products, adverse reactions, and all necessary equipment used in administering and regulating the blood products. Prior to initiating transportation, the physician will provide the paramedic with written medical orders for the treatment of any adverse reaction(s) the patient might have. When transporting the patient, at least epinephrine, Benadryl, in the patient compartment. If the transporting paramedic has not received specific training, the paramedic may refuse to transport the patient with blood or blood products infusing. The ambulance service must maintain a record of all personnel completing this specialized training.

PROCEDURE:

- Blood should be infused within a 4-hour period, otherwise there is risk of clotting in the bag.
- Carefully check blood type for compatibility with the patient BEFORE beginning the transfusion.
- Check vital signs prior to the transfusion.
- The blood should be run through at least a 20-gauge IV catheter or larger with the blood hung three to four feet above the patient.
- The IV line should be flushed with Normal Saline prior to beginning the transfusion. **Blood should be administered only with Normal Saline IV fluid.** Dextrose causes red cells to clump, swell, and hemolyze; calcium (Lactated Ringer) may cause blood to clot.
- The transfusion should be initiated at a rate of 50 mL/hr, for the first ten minutes then as ordered by the referring physician.
- Patient condition and vital signs should be monitored closely during the transfusion.
- The blood should be mixed during the transfusion by inverting the bag occasionally.

After, the transfusion is completed, flush the IV tubing until clear with Normal Saline and maintain the IV as ordered by the referring physician. If a reaction occurs during the transfusion, terminate the transfusion immediately. Initiate the treatment ordered by the transferring physician and establish medical control as soon as possible. Save the donor blood for testing at the receiving facility.

ADVERSE REACTIONS:

- Circulatory Overloading: Dyspnea, increase in blood pressure, and jugular vein distention.
- Febrile Reaction: Chilling, fever, headache, flushing, tachycardia and anxiety.
- Septic Reaction: Chilling, fever, headache, tachycardia, and hypotension.

IMMUNOLOGIC REACTION:

- Flushing, itching, rash, urticaria, and asthmatic wheezing.
- Acute Hemolytic Reaction:
 - Severe reaction which may cause back pain.
 - Dyspnea, Hypotension.
 - Diaphoresis.
 - Cold skin.
 - Jugular Vein Distention.
 - Disseminated Intravascular Coagulation.
 - Death.

IF PROBLEMS OCCUR:

- Discontinue blood administration and flush tubing with normal saline.

CALL MEDICAL CONTROL TO CONSIDER:

- Benadryl 25 mg IV, if allergic signs and symptoms are present.
- Fluids if hypotensive.

CENTRAL VASCULAR ACCESS / INDWELLING CATHETER ACCESS

PARAMEDIC

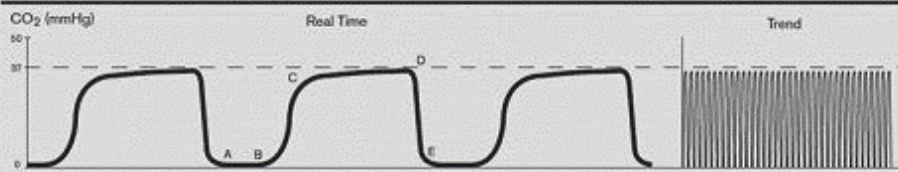
- *PICC lines or Midline caths may be accessed in unstable patients that IV/IO access cannot be rapidly achieved.*
- *Indwelling caths may be accessed in stable patients with poor IV access options or patient preference.*
- *If this access site has not been used (accessed, flushed) in the past 5 days, it should not be accessed at the time for EMS needs. The risk for clot development is too acute.*

PROCEDURE / PROTOCOL

- Always access lines utilizing needless connectors or injection caps.
 - Direct access breaches the system and increases chance of infection.
- Always cleanse the injection cap with alcohol prior to access.
 - Wipe vigorously for 30 seconds and allow to dry.
- How to assess for patency of catheter:
 - Unclamp the line, attempt to withdraw at least 10 mL of blood from the line, DISCARD that blood, do not reinfuse....
 - Attempt to flush the line with NS, using nothing smaller than a 10 mL syringe....
 - Re-clamp the line as you are completing the flush.
 - All lines should be clamped when not in use.
- Treat PICC line / Mid-Line Catheter access as you would treat saline locks.
 - Once they have been accessed and used, then make sure they are flushed with NS and then locked backed off.
- If you feel that you have breached the system for any reason:
 - Inform the patient's primary in-hospital care giver.
 - Document all actions taken.
- Document reason for PICC line / Mid-Line access.

NOTE:

- There are NO standing orders for accessing "Ports" or "Portacaths" imbedded under the skin.
- There are also NO standing orders for accessing "Vas-Cath" devices commonly seen in dialysis patients.
 - DO NOT attempt to access these devices in the pre-hospital setting.

CAPNOGRAPHY QUICK REFERENCE**Normal Capnogram****Normal EtCO₂: 35 – 45 mmHg**

The "normal" capnogram is a waveform which represents the varying CO₂ level throughout the breath cycle.

Waveform Characteristics:

A-B	Baseline	D	End-Tidal Concentration
B-C	Expiratory Upstroke	D-E	Inspiration
C-D	Expiratory Plateau		

CAUSES OF ELEVATED EtCO₂**METABOLISM**

Pain
Hyperthermia
Shivering

RESPIRATORY SYSTEM

Respiratory insufficiency
Respiratory depression
COPD
Analgesia/sedation

CIRCULATORY SYSTEM

Increased cardiac output

MEDICATIONS

Bicarbonate administration

CAUSES OF DECREASED EtCO₂**METABOLISM**

Hypothermia
Metabolic acidosis

RESPIRATORY SYSTEM

Alveolar hyperventilation
Bronchospasm
Mucus plugging

CIRCULATORY SYSTEM

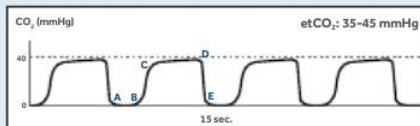
Hypotension
Sudden hypovolemia
Cardiac arrest
Pulmonary emboli

NORMAL AND ABNORMAL etCO₂/CAPNOGRAPH WAVEFORMS**Normal Capnogram**

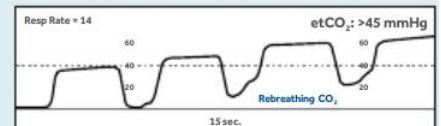
The normal capnogram is a waveform which represents the varying CO₂ level throughout the breath cycle.

Waveform Characteristics:

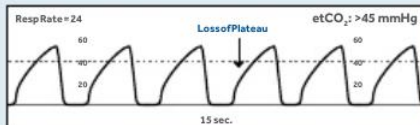
A-B: Baseline D: End-Tidal Concentration
B-C: Expiratory Upstroke D-E: Inspiration
C-D: Expiratory Plateau

**Rebreathing CO₂****Other Possible Causes:**

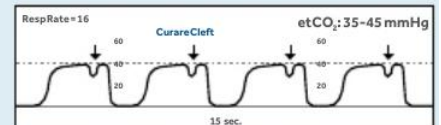
- Faulty expiratory valve
- Inadequate inspiratory flow
- Partial rebreathing
- Insufficient expiratory time

**Bronchospasm/Asthma****Other Possible Causes:**

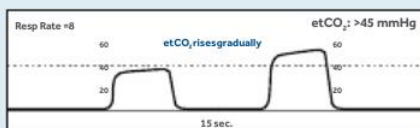
- Bronchospasm/COPD
- Obstruction in the expiratory limb of the breathing circuit
- Presence of a foreign body in the upper airway
- Partially kinked or occluded artificial airway

**Curare Cleft****Other Possible Causes:**

- Patient is mechanically ventilated
- Depth of cleft is proportional to degree of muscle relaxants

***Increasing etCO₂ (Hypoventilation)****Other Possible Causes:**

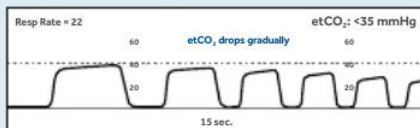
- Decrease in respiratory rate
- Decrease in tidal volume
- Increase in metabolic rate
- Rapid rise in body temperature (malignant hyperthermia)

**Cardiac Arrest****Other Possible Causes:**

- Decreased or absent cardiac output
- Decreased or absent pulmonary blood flow
- Sudden decrease in CO₂ values

***Decreasing etCO₂ (Hyperventilation)****Other Possible Causes:**

- Increase in respiratory rate
- Increase in tidal volume
- Metabolic acidosis
- Fall in body temperature

**Return of Spontaneous Circulation****Other Possible Causes:**

- Increase in cardiac output
- Increase in pulmonary blood flow
- Gradual increase in CO₂ production



*Assumes adequate circulation and alveolar gas exchange

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SPECIAL EVENT / FIRST – AID PROTOCOL (USE OF SPECIALTY ITEMS)

EMR	EMT	AEMT	PARAMEDIC
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Overview:

This protocol provides medical direction / standing orders for the usage of specialty items in the application of general patient care and first-aid. This protocol applies to patients being transported by Sumner County EMS and to any person receiving care for minor injuries during public events, stand-bys, and special operations.

Protocol for the following specialty items:

Waterjel brand burn gel - This item contains active ingredient Lidocaine HCL 2%, which is an external analgesic. This product may be used to provide temporary pain relief for superficial burns affecting less than 10% body surface area only (minor burns). If the patient was to develop rash or other suspected skin irritation, immediately wipe away with a dry cloth then flush with water.

Sting-Swabs - This item contains active ingredients Benzocaine 20% and Menthol 1%, both are topical analgesics. This product may be used to provide pain relief from insect bites and stings. Carefully monitor the patient for any indications of anaphylaxis. If the patient does not want EMS transport, complete a patient care report to include refusal of transport if the patient has any signs/symptoms indicating allergic reaction. Do not use in the eyes.

Bacitracin Ointment - This item contains active ingredient Bacitracin zinc 500 units, used as a first-aid antibiotic. This product may be used to prevent minor skin infections caused by small cuts, scrapes, or burns. Bacitracin works by stopping the growth of certain bacteria. It belongs to a class of drugs known as antibiotics. This product is intended for temporary / first-aid use only, and some wounds may require further medical treatment by a physician. The patient should be educated on the possible need for an updated tetanus shot and/or sutures for more serious soft tissue injuries. Minor wounds from cuts and scrapes should be cleansed appropriately prior to application of this product then covered with a sterile dressing/bandage.

Petroleum Gauze - This item contains a petroleum gauze strip, and may be used to dress minor wounds where clots or other fluids may adhere to regular gauze pads and dressings. This is to prevent the tissue/wound from sticking to the dressing and promote removal of the first-aid dressing when further medical attention is provided. This may also work well for other exudate (draining) wounds that require dressing/bandaging.

Hemostatic Gauze/Dressing – “Quick Clot” Hemostatic Gauze may be used to "pack" or cover heavily bleeding wounds. This product is for temporary external use to control traumatic bleeding. These specialty dressings should be avoided in minor head lacerations that seem to be bleeding heavily, yet are easily controlled with appropriate direct pressure. This is due to likely coagulation into the hair that will possibly complicate further medical treatment at the hospital.

SOG: STANDARD MEDICAL ABBREVIATIONS

Effective Date: October 1, 2016, reviewed September 10, 2025, approved for continued use.

Approved: Chief of EMS

Rationale:

To assist in maintaining accurate and consistent documentation by utilizing abbreviations, the following policy shall be followed.

Standard Operating Guideline:

When using medical abbreviations in reports, personnel may only utilize abbreviations from Sumner County Emergency Medical Service's standard medical abbreviation list. Personnel shall refrain from using abbreviations not on this list. The list may be updated from time to time, so please look at the version number and date to ensure you have the most current list.

<u>ABBREVIATION</u>	<u>DEFINITION</u>
&	AND
@	AT
X	TIMES
–	NEGATIVE
+	PRESENT, YES, OR POSITIVE
#	NUMBER
=	EQUALS
>	GREATER THAN
<	LESS THAN
%	PERCENT

Abbreviation	Meaning
<u>A's</u>	
AAO	Awake, Alert, and Oriented
ABC	Airway, Breathing, and Circulation
ABD	Abdominal
AC	Antecubital
ACLS	Advanced Cardiac Life Support
AE	Air Evac (Aeromedical)

AED	Automated External Defibrillator
AEMT	Advanced Emergency Medical Technician
AFIB	Atrial Fibrillation
AHA	American Heart Association
AICD	Automatic Internal Cardiac Defibrillator
A.I.D.S	Acquired Immune Deficiency Syndrome
AKA	Above the Knee Amputation
AMA	Against Medical Advice
AMI	Acute Myocardial Infarction
Amt	Amount
AOS	Arrival On Scene
ARDS	Adult Respiratory Distress Syndrome
ASA	Aspirin
ASAP	As soon as possible
AV	Atrioventricular
AVPU	Alert, Verbal, Painful, Unresponsive
<u>B's</u>	
BBB	Bundle Branch Block
bid	Twice a day
BKA	Below the Knee Amputation
BLS	Basic Life Support
BM	Bowel Movement
BP	Blood Pressure
BPM	Beats per minute OR Breaths per minute
BRADY	Bradycardia
BS	Blood Sugar
BVM	Bag Valve Mask
<u>C's</u>	
C1 – C7	Cervical Vertebrae 1 - 7
CA	Cancer
CABG	Coronary Artery Bypass Grafting
CAD	Coronary Artery Disease
CCEMT-P	Critical Care Paramedic
C-COLLAR	Cervical Collar
C/C	Chief Complaint
CCU	Critical Care Unit
CEBBS	Clear Equal Bilateral Breath Sounds
C.H.A.R.T	Complaint, History, Assessment, Rx, Tx
CHF	Congestive Heart Disease
CID	Cervical Immobilization Device
cm	Centimeter
CMC	Centennial Medical Center
CNS	Central Nervous System
C/O	Complaining of
COPD	Chronic Obstructive Pulmonary Disease
C.P.	Cerebral Palsy
CP	Chest Pain
CPR	Cardiopulmonary Resuscitation
CSF	Cerebrospinal Fluid
CT	Computer Aided Tomography
CTFD	Cottontown Fire Department (squad 7)
CV	Cardiovascular

SUMNER COUNTY EMERGENCY MEDICAL SERVICES PROTOCOLS

CVA	Cerebral Vascular Accident
D/C	Discontinue
<u>D's</u>	
D25	Dextrose 25%
D50	Dextrose 50%
DM	Diabetes Mellitus
DNR	Do Not Resuscitate
DOA	Dead On Arrival
DT	Delirium Tremens
<u>E's</u>	
ECG	Electrocardiogram
ED	Emergency Department
EEG	Electroencephalogram
EJ	External Jugular
EKG	Electrocardiogram
EMA	Emergency Management Agency
EMS	Emergency Medical Services
EMT	Emergency Medical Technician
EMT-IV	Emergency Medical Technician – Intravenous Therapy
EMT-P	Emergency Medical Technician – Paramedic
EOR	End of Report
ER	Emergency Room
ETA	Estimated time of arrival
etc	And so forth
ETOH	Ethly Alcohol
ETT	Endotracheal Tube
<u>F's</u>	
F	Fahrenheit
Fe	Iron
fl. oz.	Fluid Ounce
FNP	Family Nurse Practitioner
Fx	Fracture
<u>G's</u>	
GCFD	Gallatin Community Fire Department (squad 6)
GFD	Gallatin Fire Department
GI	Gastrointestinal
gm	Gram
GOA	Gone on arrival
GPD	Gallatin Police Department
Gravida	Number of Pregnancies
GSW	Gunshot Wound
GTT	Drop
GVFD	Goodlettsville Fire Department
GVPD	Goodlettsville Police Department
Gyn	Gynecology
<u>H's</u>	
H	Hour
HA	Headache
HBV	Hepatitis B
HEENT	Head, Eyes, Ears, Nose, and Throat.
HFD	Hendersonville Fire Department
HIFD	Highland Fire Department (squad 1)

SUMNER COUNTY EMERGENCY MEDICAL SERVICES PROTOCOLS

HIV	Human Immunodeficiency Virus
HMC	Hendersonville Medical Center
HPD	Hendersonville Police Department
HR	Heart Rate
HTN	Hypertension
HX	History
<u>I's</u>	
IC	Incident Command
ICP	Intracranial Pressure
ICS	Intracostal Space
ICU	Intensive Care Unit
ID	Intradermal
IDDM	Insulin Dependent Diabetes Mellitus
IM	Intramuscular
IO	Intraosseous
IV	Intravenous
IVP	Intravenous Push
<u>J's</u>	
JVD	Jugular Vein Distention
K.E.D.	Kendrick Extrication Device
<u>K's</u>	
K+	Potassium
KCL	Potassium Chloride
Kg	Kilogram
KVO	Keep Vein Open
<u>L's</u>	
L1 – L5	Lumbar Vertebrae 1 - 5
L	Left
L&D	Labor and Delivery
Lab	Laboratory
LAD	Left Anterior Descending Artery
LBBB	Left Bundle Branch Block
Lbs.	Pounds
LF	Life Flight (Aeormedical)
Lg	Large
LLE	Left Lower Extremity
LLL	Left Lower Lobe
LLQ	Left Lower Quadrant
LMP	Last Menstrual Period
LOC	Level of Consciousness OR Loss of Consciousness
lpm	Liters Per Minute
LPN	Licensed Practical Nurse
LR	Lactated Ringer
LRS	Lactated Ringer Solution
LSB	Long Spine Board
LUE	Left Upper Extremity
LUL	Left Upper Lobe
LUQ	Left Upper Quadrant
<u>M's</u>	
m	Meter
MAEW	Moves All Extremities Well
mcg	Microgram

SUMNER COUNTY EMERGENCY MEDICAL SERVICES PROTOCOLS

MCL	Mid-clavicular Line
M.D.	Medical Director
MD1	Medical Director (Ray Pinkston, MD)
MD2	Medical Director (Duane Harrison MD)
ME	Medical Examiner
MEDS	Medication
mEq	Milliequivalent
MFD	Millersville Fire Department
mg	Milligram
MGSO4	Magnesium Sulfate
MI	Myocardial Infarction
mL	Milliliter
mm	Millimeter
MM	Mile Marker
MPD	Millersville Police Department
MR	Mental Retardation
MRI	Magnetic Resonance Imaging
MS	Multiple Sclerosis
MVA	Motor Vehicle Accident
MVC	Motor Vehicle Collision
<u>N's</u>	
N/A	Not Applicable
N/V	Nausea / Vomiting
N/V/D	Nausea / Vomiting / Diarrhea
NAD	No Apparent Distress
NC	Nasal Cannula
NEURO	Neurological
NFD	Nashville Fire Department
NGT	Nasogastric Tube
NIDDM	Non-insulin Dependent Diabetes Mellitus
NKDA	No Known Drug Allergies
NOFD	Number One Fire Department (squad 12)
NRB	Non Rebreather
NS	Normal Saline
NSR	Normal Sinus Rhythm
NTG	Nitroglycerin
<u>O's</u>	
O2	Oxygen
OB	Obstetric
OGFD	Oak Grove Fire Department (squad 8)
OTC	Over The Counter
Oz	Ounce
<u>P's</u>	
PA	Physician Assistant
PAC	Premature Atrial Contraction
PARA	Number of Live Births
PAT	Pediatric Assessment Triangle
PCI	Percutaneous Coronary Intervention – Cardiac Cath
PCO2	Partial Pressure of Carbon Dioxide
PE	Pulmonary Embolism
PED	Pediatric
PERRL	Pupils, Equal, Round, and Reactive to Light

SUMNER COUNTY EMERGENCY MEDICAL SERVICES PROTOCOLS

PFD	Portland Fire Department
PID	Pelvic Inflammatory Disease
PJC	Premature Junctional Contraction
PMC-ER	Portland Medical Center – Emergency Room
P.M.	Afternoon and Evening
PMHX	Past Medical History
PMS	Pulse, Movement, and Sensation
p.o.	By mouth
pO2 or PO2	Partial Pressure of Oxygen
POC	Position of Comfort
POV	Personal Owned Vehicle
PPD	Portland Police Department
PRN	As Needed
PSVT	Paroxysmal Supraventricular Tachycardia
PT	Patient
PTA	Prior to Arrival
PTD	Prior to Departure
PUTS	Patient Unable to Sign
PVC	Premature Ventricular Contraction
PVD	Peripheral Vascular Disease
P/W/D	Pink, Warm, and Dry
<u>Q's</u>	
q	Every
qt	Quart
<u>R's</u>	
R	Right
RBBB	Right Bundle Branch Block
RLE	Right Lower Extremity
RLL	Right Lower Lobe
RLQ	Right Lower Quadrant
RML	Right Middle Lobe
RN	Registered Nurse
R/O	Rule Out
ROM	Range of Motion
RT	Respiratory Therapy
RUE	Right Upper Extremity
RUL	Right Upper Lobe
RUQ	Right Upper Quadrant
RX	Treatment, Prescribed for, Therapy
<u>S's</u>	
S1 – S5	Sacrum Vertebrae 1 - 5
SAMPLE	S/S, Allergies, Medications, Past Medical Hx, Last Oral Intake, Events Leading Up To
SaO2	Oxygen Saturation by blood analysis (blood gas)
SCSO	Sumner County Sheriff Office
SEFD	South East Fire Department (squad 3)
SIFD	Shackle Island Fire Department (squad 11)
SL	Sublingual
SLMC	Skyline Medical Center
SMC	Summit Medical Center
SOB	Short of Breath
SQ	Subcutaneous
SRMC	Sumner Regional Medical Center

SUMNER COUNTY EMERGENCY MEDICAL SERVICES PROTOCOLS

SRO	School Resource Officer
S/S	Signs and Symptoms
ST	Sinus Tachycardia
STEMI	ST Elevation Myocardial Infarction
STMT	St. Thomas Mid-town Hospital (formerly Baptist)
STW	St. Thomas West Hospital
SVT	Supraventricular Tachycardia
<u>T's</u>	
T1 - T12	Thoracic Vertebrae 1 - 12
TBI	Traumatic Brain Injury
THP	Tennessee Highway Patrol
TIA	Transient Ischemic Attack
t.i.d.	Three Times A Day
TKO	To Keep Open
tPA	Tissue Plasminogen Activator
TSI	Total Spinal Immobilization
TVI	Total Volume Infused
Tx	Treatment
<u>U's</u>	
U	Unit
UMC	University Medical Center
UOA	Upon Our Arrival
URI	Upper Respiratory Infection
UTI	Urinary Tract Infection
<u>V's</u>	
VA	Veteran Affairs
VF	Ventricular Fibrillation
VT	Ventricular Tachycardia
VUMC	Vanderbilt University Medical Center
<u>W's</u>	
WCFD	White House Community Fire Department (51)
WFD	Westmoreland Fire Department
WHFD	White House Fire Department (City)
WHPD	White House Police Department
WNL	Within Normal Limits
WPD	Westmoreland Police Department
Wt	Weight
<u>X,Y,Z's</u>	N/A

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- Edited and updated by Sumner County EMS Protocol Review Committee with approval from Medical Director Maddison Gowder, MD